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Doble Grado en Física y Matemáticas

Quantum Computation & Quantum Information

Comprehensive Lecture Notes and Algorithm Analysis

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Preface

These lecture notes constitute a comprehensive, mathematically rigorous synthesis of the *Quantum Computation and Quantum Information* course, originally taught by Professor Ryan O'Donnell at Carnegie Mellon University (CMU). The core curriculum, conceptual frameworks, and original video lectures are publicly available educational resources.

While the foundational physics, theoretical curation, and structural assembly are the result of dedicated human study, the transcription processing, LaTeX typesetting, and vector graphic generation within this document were supported by Artificial Intelligence tools to ensure typographical precision.

These notes are intended as a formal study guide, a repository of algorithmic analysis, and a tribute to the pedagogical excellence of the original CMU course.

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1 Lecture 1: Introduction to Quantum Computation

Quantum computation and quantum information are not isolated domains but reside at the exact intersection of three fundamental disciplines: Physics, Mathematics, and Computer Science. While the laws governing the behavior of quantum computers come from quantum mechanics, their rigorous study and the evaluation of their computational power are formulated through algebraic axioms and complexity theory.

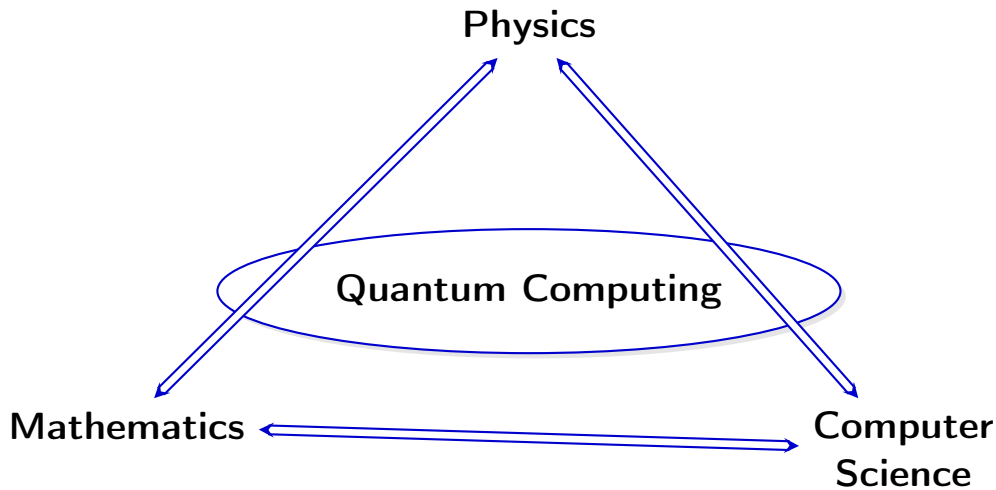


Figure 1: Foundational pillars of Quantum Computing.

The central theme structuring this discipline is **computational complexity**. The fundamental question is whether quantum computers are inherently more “powerful” than their classical counterparts and, if so, for which subclass of problems they manifest an empirical or theoretical advantage.

1.1 The Limits of Classical Computing: Physical Numbers

To understand the impact of quantum computing, we must first establish a rigorous distinction between magnitudes that are tractable in our universe and those that are purely abstract.

- **Physical Numbers:** Those that can quantify tangible objects or processable events within the thermodynamic and spatiotemporal constraints of our universe.
- **Unphysical Numbers:** Quantities that vastly exceed the material capacity of the cosmos.

To put this into perspective, consider the following scaling hierarchy of classical computing resources:

$$\begin{aligned}
 10^9 &\approx 2^{30} && \text{(Instructions per second on a standard processor)} \\
 10^{12} &\approx 2^{40} && \text{(FLOPS on a modern gaming console)} \\
 2^{50} &&& \text{(Storage scale in massive data centers)} \\
 2^{64} &&& \text{(Memory address limit in 64-bit architectures)}
 \end{aligned}$$

Even when extrapolating these resources to a cosmological scale, there is a strict thermodynamic upper bound:

$$10^{80} \approx \text{Estimated number of elementary particles in the observable universe} \quad (1)$$

Any computational requirement that significantly exceeds 10^{80} operations (for example, an algorithm requiring 10^{500} steps) belongs to the domain of unphysical numbers. Its execution is materially impossible, regardless of future advances in classical engineering.

1.2 The Cost of Cryptography: Integer Multiplication vs. Factorization

The asymmetry in the computational difficulty of certain inverse mathematical operations constitutes the foundational security pillar of contemporary public-key cryptography (such as the RSA cryptosystem). Let us analyze these two inverse problems for integers of length $N = 500$ digits.

Problem 1: Integer Multiplication In the elementary classical multiplication algorithm, the number of elementary operations $T(N)$ scales quadratically with the number of input digits:

$$T(N) \propto N^2 \implies T(500) \approx 250,000 \text{ steps} \quad (2)$$

Since N^2 is a polynomial function of the input size, this is classified as a **Polynomial (P) algorithm** (a physical or tractable algorithm), which is easily executed by classical hardware. (Note: Utilizing the classical Fast Fourier Transform (FFT), this asymptotic time complexity can be further reduced to $\mathcal{O}(N \log N)$).

Problem 2: Integer Factorization Given a composite 500-digit integer, finding its prime factors using trial division or brute force requires an execution time that scales exponentially:

$$T(N) \propto C^N \quad (\text{Non-polynomial}) \quad (3)$$

For $N = 500$, the number of necessary operations scales up to approximately 10^{250} steps. Given that $10^{250} \gg 10^{80}$, we are dealing with an **unphysical number**. No classical computer can solve this within the lifespan of the universe. Even advanced modern subexponential algorithms, such as the General Number Field Sieve (GNFS) which runs in $\approx \mathcal{O}\left(e^{cN^{1/3}(\log N)^{2/3}}\right)$, fail to achieve polynomial time.

1.3 The Quantum Paradigm and Shor's Algorithm

In 1994, Peter Shor rigorously demonstrated the existence of a quantum algorithm capable of factoring integers in polynomial time:

$$T_{\text{Shor}}(N) \approx \mathcal{O}(N^2) \text{ steps} \quad (4)$$

This theoretical breakthrough implies that a large-scale quantum computer can break RSA cryptography in a practical time frame. The immense power of this algorithm stems directly from a fundamental axiom of quantum mechanics: the state space.

If we possess 1,000 classical two-level particles, classical mechanics describes their joint state using a polynomial number of parameters. However, quantum mechanics dictates that the joint state $|\psi\rangle$ of 1,000 interacting two-level systems (*qubits*) is a coherent superposition over all possible configurations simultaneously. Mathematically, this requires a linear combination of 2^{1000} complex probability amplitudes:

$$|\psi\rangle = \sum_{x=0}^{2^{1000}-1} \alpha_x |x\rangle \quad \text{where} \quad \sum_x |\alpha_x|^2 = 1 \quad \text{and} \quad \alpha_x \in \mathbb{C} \quad (5)$$

Nature inherently maintains an exact mathematical record of these 2^{1000} numbers. Quantum computation is, in essence, a method to *hack* this massive natural state register. By exploiting the Quantum Fourier Transform (QFT), the algorithm induces quantum interference across these 2^{1000} amplitudes, exponentially amplifying the probability of measuring the correct physical result.

Ontological Interpretation: The necessity of operating simultaneously on 2^{1000} amplitudes raises profound physical questions. The Many-Worlds Interpretation, formulated by Hugh Everett III, suggests that this vast amount of information is processed by distributing the computation across 10^{500} interacting parallel universes. While highly vivid and conceptually useful, this philosophical framework is not strictly necessary to formulate the underlying mathematical physics; it suffices to assume the formal axioms of Hilbert space and the unitary evolution dictated by linear quantum mechanics.

2 Lecture 2: Rotate, Compute, Rotate

2.1 Quantum Mechanics: Probability with Minus Signs

The foundational laws governing the behavior of subatomic particles can be elegantly summarized by a single, counterintuitive phrase: quantum mechanics is probability with minus signs. If one were tasked with inventing a conservative mathematical extension of classical probability theory where probabilities were permitted to be negative numbers, the resulting framework would inevitably be quantum mechanics. This perspective allows us to understand quantum computing not as an alien paradigm, but as a direct extension of classical computing endowed with one extraordinary, non-classical power.

2.2 Probabilistic vs. Deterministic Computing

To formalize this transition to quantum logic, we must first rigorously examine how classical probabilistic computing differs from strict deterministic computing. Consider augmenting standard deterministic architectures—such as boolean circuits or standard Turing machines—with a single hardware instruction: a coin flip. By definition, this instruction evaluates to 0 with a probability of $1/2$ and 1 with a probability of $1/2$.

Is a probabilistic computer fundamentally more powerful than a deterministic one? If the computational task inherently requires simulating random physical phenomena or generating random cryptographic keys, the probabilistic machine is trivially superior by definition. However, when restricted strictly to computing fixed deterministic functions (e.g., multiplying two numbers, testing primality, or finding minimum spanning trees), the advantage is less obvious.

In functional computation, the primary theoretical merit of probabilistic algorithms lies in trading a mathematically negligible probability of error for a significant increase in algorithmic efficiency. A quintessential historical example is the problem of primality testing for an n -bit integer (e.g., $n = 1024$).

The historical progression of the time complexity for primality testing algorithms illustrates this trade-off explicitly:

$$\begin{aligned}
 T_{\text{Miller, 1976}}(n) &= \mathcal{O}(n^4) \text{ steps} && \text{(Deterministic, assuming Extended Riemann Hypothesis)} \\
 T_{\text{Solovay-Strassen, 1977}}(n) &= \mathcal{O}(n^3) \text{ steps} && \text{(Probabilistic, bounded error)} \\
 T_{\text{Miller-Rabin, 1980}}(n) &= \mathcal{O}(n^2) \text{ steps} && \text{(Probabilistic, unconditional)} \\
 T_{\text{AKS, 2002}}(n) &= \mathcal{O}(n^{12}) \text{ steps} && \text{(Deterministic, unconditionally proven)} \\
 T_{\text{Lenstra-Pomerance}}(n) &= \mathcal{O}(n^6) \text{ steps} && \text{(Deterministic, optimized)}
 \end{aligned}$$

Despite the explicit polynomial speedups observed in these methods, complexity theorists possess strong theoretical evidence leading to a foundational conjecture: every problem solvable in polynomial time probabilistically (the class BPP) is also solvable in polynomial time deterministically (the class P). Consequently, probabilistic computation is strongly believed to provide polynomial speedups for functional tasks, but it does not yield exponential speedups.

2.3 State Representation and Exponential Scaling

To bridge the gap to quantum mechanics, let us analyze the mathematical state space of a classical probabilistic algorithm. Consider the following classical initialization sequence:

```
Initialize array A of length 1000
for 0 <= i <= 1000:
    A[i] := CoinFlip(0/1)
```

Upon execution of this loop, the state of the memory array A is not defined by 1000 deterministic bits, but rather by the 2^{1000} probabilities corresponding to every possible binary sequence. In classical systems, these 2^{1000} numbers merely represent the observer's lack of knowledge regarding the actual physical state of the memory.

In stark contrast, a quantum computer physically embodies this vast state space. If we initialize an array of 1000 photons (qubits) and pass them through a precise physical obstacle course composed of mirrors, lasers, and prisms, the true physical state of the system at the output is completely characterized by 2^{1000} complex amplitudes. Unlike classical probabilities, these amplitudes represent an ontologically real, simultaneous superposition of all possible configurations.

2.4 The Extra Power of Quantum Computing: The Discrete Fourier Transform

What concrete computational capability does this exponential amplitude space grant us? It provides the unprecedented ability to detect global patterns within implicitly represented data sets.

Because an array of 2^{1000} entries cannot physically exist in our universe, the data list must be implicitly generated by a computable function. We construct a boolean circuit that computes $f : \text{domain} \rightarrow \mathbb{C}$, evaluating the x -th entry of the list in polynomial time. The quantum computer computes this function in superposition, embedding the 2^{1000} values of $f(x)$ into the amplitudes of the 1000 qubits.

To extract the global patterns of $f(x)$, the quantum computer applies the Discrete Fourier Transform (DFT). The mathematical operation of the Fourier transform depends heavily on the topology of the chosen domain:

1. **Continuous Domain ($\mathbb{R}$ or $[0, 2\pi]$):** The standard Fourier transform decomposes continuous signals into a basis of continuous sines and cosines.
2. **Discrete Integers Modulo N ($\mathbb{Z}_N$ where $N = 2^n$):** The standard DFT discretizes the basis functions into complex roots of unity.
3. **The Boolean Cube ($\{0, 1\}^n$):** The Walsh-Hadamard transform decomposes boolean functions into a basis of XOR functions.

Let $N = 2^n$ be the dimension of the vector space, corresponding to an n -qubit system. Applying the Fourier transform to a signal requires multiplying the N -dimensional state vector by the $N \times N$ discrete Fourier matrix. The scaling of computational resources required for this transformation highlights the source of quantum computational advantage:

$$\begin{aligned} T_{\text{Naive}}(N) &= \mathcal{O}(N^2) \text{ steps} \\ T_{\text{FFT}}(N) &= \mathcal{O}(N \log N) \text{ steps} \quad (\text{Cooley-Tukey classical FFT}) \\ T_{\text{Quantum}}(N) &= \mathcal{O}(\log N) \text{ steps} \quad (\text{Quantum Fourier Transform}) \end{aligned}$$

The Quantum Fourier Transform operates exponentially faster than the best classical algorithm, requiring only $n = \log_2 N$ computational steps to globally interfere 2^n amplitudes.

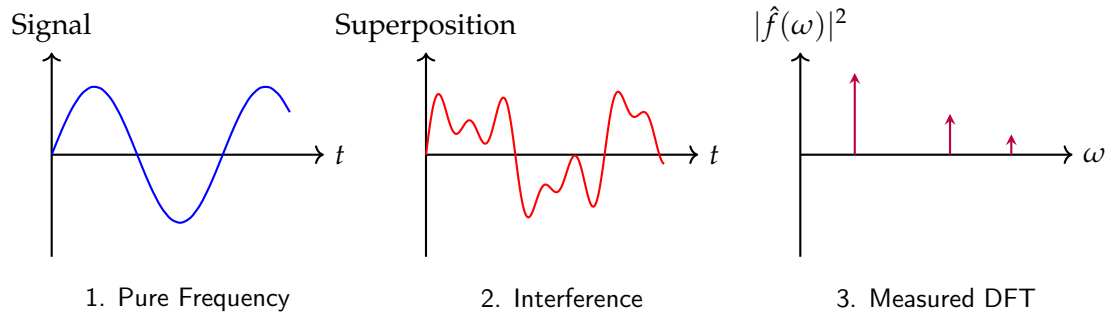


Figure 2: Visual intuition of the Fourier Transform extracting discrete global patterns from superposed states.

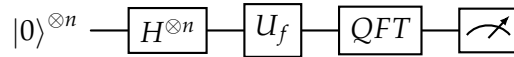


Figure 3: The standard architecture of exponential quantum speedups: Superposition, Oracle computation, Interference via QFT, and Measurement.

2.5 Quantum Measurement and State Collapse

Crucially, the mathematics of quantum mechanics prohibits the direct observation or readout of the 2^{1000} resulting Fourier coefficients, denoted as $\hat{f}(y)$. The quantum extraction of data relies entirely on the Born rule of measurement.

When the n -qubit register is passed through a measurement apparatus (such as a photon detector), the system probabilistically outputs a single n -bit string y^* . The probability $P(y^*)$ of detecting this specific frequency component y^* is strictly given by the squared magnitude of its complex amplitude:

$$P(y^*) = |\hat{f}(y^*)|^2 \quad (6)$$

Because the unitary operation preserves the norm of the state vector, the sum of all squared amplitudes equals 1, ensuring that this forms a valid probability distribution. Immediately upon this observation, the massive superposition is irreversibly destroyed. The state of the system snaps into a trivial deterministic basis state corresponding to the measurement outcome:

$$|\psi_{\text{post-measurement}}\rangle = |y^*\rangle \quad (7)$$

To obtain a second independent sample from the Fourier distribution $|\hat{f}(y)|^2$, one cannot re-measure the collapsed particles; the entire quantum obstacle course must be initialized and executed from the beginning.

Ultimately, this sequence defines the unifying architecture of all exponential quantum speedups: initialize a superposition, compute an implicit function, rotate the vector space using the Fourier transform, and measure the result.

3 Lecture 3: The Formalism of Qubits and Quantum Measurement

3.1 Physical and Logical Bits

In classical computer science, the fundamental logical concept is the bit, which deterministically holds a value of either 0 or 1. Physically, this logical abstraction is implemented through macroscopic systems, such as a low or high voltage coursing through a wire, or a magnetized

region on a hard drive. These implementations typically rely on millions of physical particles flowing simultaneously to represent a single logical state.

To push computational miniaturization to its physical limits, one naturally asks whether a logical bit can be encoded into the state of a single fundamental particle, such as an electron or a photon. The behavior of single particles is governed entirely by the laws of quantum mechanics. Different physical properties of these particles can be harnessed to define our two computational states: the ground and excited orbital states of a hydrogen atom, the spin of an electron (spin-up or spin-down), or the linear polarization of a photon (horizontal or vertical).

We define a physical *measuring device* as an apparatus that probes this property and outputs a deterministic classical bit. For instance, a photon's polarization can be measured by passing it through a macroscopic filter (such as a Polaroid sheet from 3D glasses), which acts as a measurement box whose readout displays either "horizontal" or "vertical".

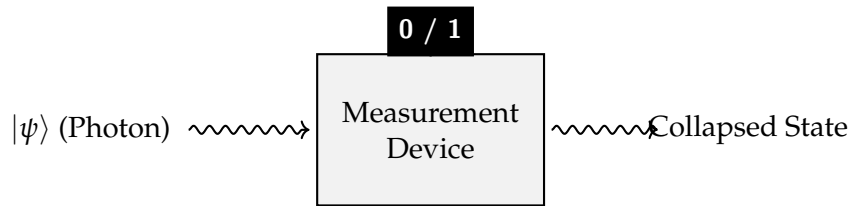


Figure 4: Abstract schema of a quantum measuring device interacting with a single particle. The device forces the superposition to resolve into a classical bit output on the digital readout.

3.2 Axioms of Quantum Mechanics: Superposition and Measurement

The transition from classical to quantum computing is governed by two foundational axioms regarding how nature preserves and extracts information.

Law 1: Quantum Superposition. If a quantum system can be in one of two basic, mutually exclusive states (which we computationally label as $|0\rangle$ and $|1\rangle$), it can also exist in a superposition of these states. Mathematically, such a state is called a *qubit* (quantum bit) and is described by:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (8)$$

where $\alpha, \beta \in \mathbb{C}$ are complex numbers known as *probability amplitudes*. These amplitudes must satisfy a strict normalization equation:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (9)$$

This framework permits an infinite continuum of valid states. For example, a valid state for a photon could be $0.8 |0\rangle + 0.6 |1\rangle$, since $0.8^2 + 0.6^2 = 0.64 + 0.36 = 1$. The amplitudes can be negative, yielding a distinct state such as $0.8 |0\rangle - 0.6 |1\rangle$, or even purely imaginary, such as $i |0\rangle + 0 |1\rangle$.

Law 2: Measurement and Wave-function Collapse. Measurement in quantum mechanics is inherently probabilistic and destructive. If a qubit in the general state $\alpha |0\rangle + \beta |1\rangle$ is passed through a measuring device configured to distinguish between 0 and 1, the physical readout will exhibit:

- Outcome 0 with probability $|\alpha|^2$
- Outcome 1 with probability $|\beta|^2$

Crucially, the measurement dynamically alters the particle itself. Upon observation, the superposition is destroyed, and the state of the particle undergoes an irreversible collapse. If the

measurement apparatus outputs 0, the physical state of the qubit instantaneously snaps to $|0\rangle$. If the apparatus reads 1, the state collapses to $|1\rangle$.

This leads to the *Quantum Zeno Effect*: if a particle is measured to be 0, any subsequent chained measurements of that identical particle in the same basis will consistently yield 0 with 100% probability, because the post-measurement amplitude on $|0\rangle$ is rigidly 1 (and $1^2 = 1$).

3.3 Generalization to Higher Dimensions: Qudits

While the canonical quantum computer operates on two-level systems (qubits), particles can exhibit a larger number of basic states. A hydrogen atom's energy levels or the spin configurations of a deuterium nucleus can hold three distinct basic states, creating a system known as a *qutrit* ($d = 3$).

For an arbitrary integer d , the generic system is called a *qudit*. Both Law 1 and Law 2 intuitively extend to dimension d . A general qudit is described by d complex amplitudes:

$$|\psi\rangle = \alpha_1 |1\rangle + \alpha_2 |2\rangle + \cdots + \alpha_d |d\rangle \quad (10)$$

subject to the normalization condition $\sum_{i=1}^d |\alpha_i|^2 = 1$. Upon measurement, the probability of obtaining the classical outcome i is exactly $|\alpha_i|^2$, forcing the state vector to instantaneously collapse completely onto the basis state $|i\rangle$.

3.4 Linear Algebra and Dirac Bra-Ket Formalism

To formalize the calculus of quantum states, we represent amplitudes as elements of vector spaces. The state of a d -level qudit maps directly to a d -dimensional complex column vector $\mathbf{v} \in \mathbb{C}^d$. The normalization constraint fundamentally means that all valid quantum states are *unit vectors* (vectors of geometric length 1) residing on the surface of a d -dimensional complex hypersphere.

To eliminate cumbersome standard matrix notation, the quantum computation community employs Dirac's bra-ket notation. The symbol $|\psi\rangle$ (pronounced "ket") declares that ψ is a column vector. For a standard qubit ($d = 2$), we replace the classical standard basis vectors (e_1 and e_2) with $|0\rangle$ and $|1\rangle$:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (11)$$

Thus, a generalized qubit state is written as a linear combination of basis kets:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (12)$$

The dual of the ket is the "bra", denoted $\langle\psi|$. The bra represents the conjugate transpose (or Hermitian adjoint, denoted by $\dagger$) of the column vector:

$$\langle\psi| = |\psi\rangle^\dagger = (\alpha^* \quad \beta^*) \quad (13)$$

The inner product of two quantum states $|u\rangle$ and $|v\rangle$ is gracefully written as the closed bracket $\langle u|v\rangle$, representing the matrix multiplication of the row vector $\langle u|$ and the column vector $|v\rangle$. Geometrically, the inner product $\langle u|v\rangle$ quantifies the magnitude of the projection of vector $|v\rangle$ onto vector $|u\rangle$. For real unit vectors, this inner product is precisely the cosine of the angle θ between them.

3.5 Geometric Representation and Change of Basis

While amplitudes exist strictly in $\mathbb{C}^2$, the fact that four real parameters are constrained by a normalization equation and a global phase equivalence allows us to simplify the visualization by pretending the amplitudes are exclusively real numbers. Under this simplification, a qubit state can be depicted as a point on the unit circle in a two-dimensional Euclidean plane.

A generic real qubit $|\psi\rangle$ characterized by an angle θ from the horizontal axis has its amplitudes uniquely defined by basic trigonometry:

$$|\psi\rangle = \cos(\theta) |0\rangle + \sin(\theta) |1\rangle \quad (14)$$

If we measure $|\psi\rangle$ in the standard computational basis $\{|0\rangle, |1\rangle\}$, the probability of obtaining 0 is the squared magnitude of its projection onto $|0\rangle$, which equals $\cos^2(\theta)$. The probability of obtaining 1 is exactly $\sin^2(\theta)$.

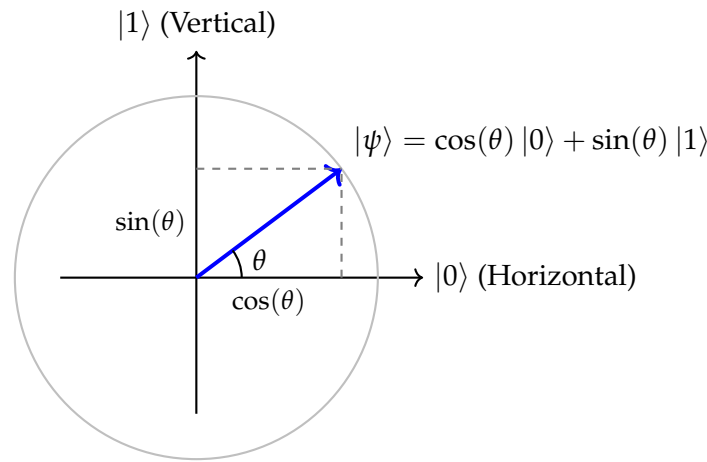


Figure 5: Geometric representation of a real-amplitude qubit mapped onto the 2D unit circle, demonstrating the amplitudes as projections along the standard computational basis vectors.

Beyond the standard basis, two particularly famous orthogonal vectors play a dominant role in quantum algorithm design: the $|+\rangle$ and $|-\rangle$ states. These represent vectors located exactly at 45° ($\pi/4$ radians) and -45° ($-\pi/4$ radians) respectively:

$$|+\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle, \quad |-\rangle = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \quad (15)$$

If we inject the $|+\rangle$ state into a standard $|0\rangle / |1\rangle$ measuring device, the probability of obtaining 0 is $|1/\sqrt{2}|^2 = 1/2$ (50%), and the probability of obtaining 1 is similarly $1/2$. A standard measuring apparatus cannot distinguish whether the input was $|+\rangle$ or $|-\rangle$ since both yield an identical 50/50 probabilistic distribution.

3.6 Measurements in Arbitrary Bases

Quantum mechanics permits measurements across any arbitrary orthogonal coordinate system. For any pair of mutually orthogonal unit vectors $\{|u\rangle, |v\rangle\}$, a dedicated physical measuring device can be engineered to probe the system along these new axes.

When a qubit $|\psi\rangle$ enters a device calibrated for the $\{|u\rangle, |v\rangle\}$ basis, the internal readout mechanism will output the classical label " u " with a probability equivalent to the squared projection of $|\psi\rangle$ onto $|u\rangle$, rigorously defined by the inner product $|\langle u|\psi\rangle|^2$. Accordingly, the outcome " v " triggers with probability $|\langle v|\psi\rangle|^2$. In tandem with Law 2, the post-measurement state strictly collapses into the newly corresponding basis vector ($|u\rangle$ or $|v\rangle$).

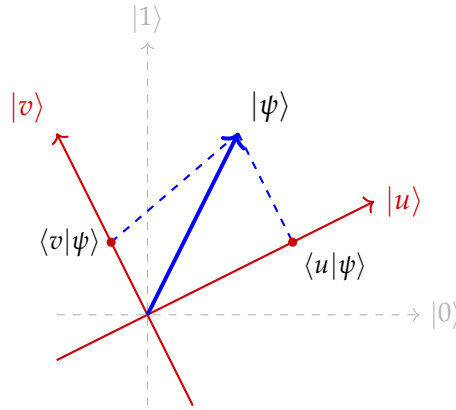


Figure 6: Measurement of the state $|\psi\rangle$ with respect to an arbitrary orthonormal basis $\{|u\rangle, |v\rangle\}$. The amplitudes dictating the probabilities are structurally identical to the inner products $\langle u|\psi\rangle$ and $\langle v|\psi\rangle$.

This principle explains counterintuitive physical phenomena, such as inserting a diagonal (45°) polarizing filter between perpendicular horizontal and vertical filters. Ordinarily, perpendicular filters block 100% of incoming light. However, injecting the 45° filter forces the incoming horizontal photons to be re-measured into the $|+\rangle / |-\rangle$ basis. This wave-function collapse realigns a portion of the photons geometrically, granting them a non-zero probability amplitude to successfully penetrate the final vertical filter, thereby creating transmission where classically none should exist.

4 Lecture 4: Unitary Transformations and the Elitzur-Vaidman Bomb

4.1 Measurements in Arbitrary Orthonormal Bases

The standard framework of quantum measurement extracts a classical bit of information from a quantum state by projecting it onto the computational basis $\{|0\rangle, |1\rangle\}$. However, physical reality permits the construction of a measuring device corresponding to any arbitrary orthonormal basis of a d -dimensional state space. Let $\{|u_1\rangle, \dots, |u_d\rangle\}$ be such a basis.

When an arbitrary state $|\psi\rangle$ is subjected to a measurement in this basis, the device outputs the classical label corresponding to $|u_i\rangle$ with a probability strictly given by the squared magnitude of its complex amplitude along that basis vector. Mathematically, this amplitude is the inner product (or projection) between the state and the basis vector:

$$P(i) = |\langle u_i|\psi\rangle|^2$$

If we restrict our analysis to states with strictly real amplitudes, the inner product $\langle u_i|\psi\rangle$ is precisely the cosine of the geometric angle θ between the two vectors. Thus, the probability of measuring state i simplifies to:

$$P(i) = \cos^2(\theta)$$

A canonical example of an alternative basis in a two-dimensional Hilbert space (a qubit) is the diagonal basis, consisting of the orthogonal vectors $|+\rangle$ and $|-\rangle$:

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \\ |-\rangle &= \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \end{aligned}$$

If we pass a particle in the state $|+\rangle$ into a device calibrated to measure in the $\{|+\rangle, |-\rangle\}$ basis, the readout will be "+" with 100% probability. Conversely, if we inject the standard computational state $|0\rangle$ into this diagonal measuring device, the symmetry of the vectors yields:

$$P(+)=|\langle +|0\rangle|^2=\left(\frac{1}{\sqrt{2}}\right)^2=\frac{1}{2}$$

$$P(-)=|\langle -|0\rangle|^2=\left(\frac{1}{\sqrt{2}}\right)^2=\frac{1}{2}$$

The measurement induces a completely random 50/50 readout, and the post-measurement wave-function collapses irreversibly into either $|+\rangle$ or $|-\rangle$ accordingly.

4.2 Physical Implementation: Polarizing Filters

To physically ground these concepts, consider the transmission of photons through polarizing lenses (such as those in 3D glasses). A polarizing filter acts simultaneously as a quantum measurement device and a state-conditional filter. A horizontal filter measures a photon in the standard basis $\{|0\rangle, |1\rangle\}$. If the outcome is $|0\rangle$ (horizontal), the photon passes through. If the outcome is $|1\rangle$ (vertical), the photon is absorbed and dissipated as heat.

Consider an experimental apparatus with three cascaded filters:

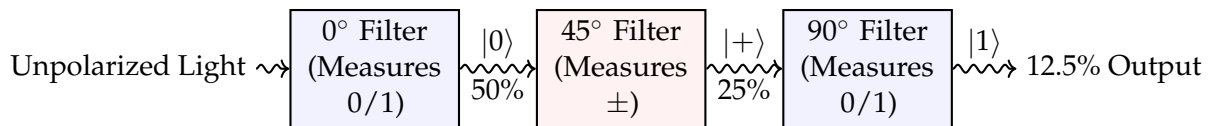


Figure 7: Three-stage polarizing filter experiment. The intermediate 45° filter rotates the measurement basis, preventing the total annihilation of the beam that would occur with strictly orthogonal filters.

If only the horizontal and vertical filters are present, 0% of the initial light traverses the setup. However, interposing a 45° diagonal filter between them forces the horizontally polarized photons ($|0\rangle$) to be measured in the $\{|+\rangle, |-\rangle\}$ basis. Half of these photons collapse into $|+\rangle$, passing through the diagonal filter. Subsequently, when these $|+\rangle$ photons encounter the vertical filter (a standard basis measurement), they again have a 50% probability of collapsing into $|1\rangle$ and passing through. The sequence paradoxically resurrects 12.5% of the original beam intensity strictly through chained quantum basis transformations.

4.3 The Elitzur-Vaidman Bomb Tester

This dynamic of measurement-induced state collapse leads directly to one of the most conceptually striking protocols in quantum information: Interaction-Free Measurement, formalized by the Elitzur-Vaidman bomb experiment (1993).

Suppose we are presented with an opaque box. Inside, it either contains nothing (a "dud") or an explosive device equipped with a highly sensitive horizontal polarizing filter. The bomb filter measures incident photons in the standard basis $\{|0\rangle, |1\rangle\}$. If it measures $|0\rangle$, the photon passes uninterrupted. If it measures $|1\rangle$, the photon is absorbed, triggering an immediate and catastrophic explosion.

Our task is to determine whether the box contains a bomb without detonating it. Under classical physics, this task is fundamentally impossible. Sending a $|0\rangle$ photon yields no information, as it passes safely through both a dud and a bomb. Sending a $|1\rangle$ photon perfectly identifies the bomb but inevitably causes an explosion.

4.3.1 Initial Quantum Strategy

A naive quantum strategy injects a photon in the superposition state $|+\rangle$. Inside a bomb box, the measurement collapses the state:

- With 50% probability, it collapses to $|1\rangle$, resulting in an explosion.
- With 50% probability, it collapses to $|0\rangle$, passing safely.

If the photon emerges safely, we pass it through a final diagonal filter measuring in the $\{|+\rangle, |-\rangle\}$ basis. If the box was a dud, the emergent photon remains $|+\rangle$, guaranteeing a + readout. If the box contained a bomb, the emergent photon collapsed to $|0\rangle$. Measuring $|0\rangle$ in the diagonal basis yields a – readout with 50% probability.

Thus, receiving a – readout guarantees the presence of a bomb without having triggered it. This provides a 25% success rate. While profound, a 50% mortality rate remains unacceptable for practical applications.

4.3.2 The High-Probability Zeno Protocol

By leveraging an iterative series of minute state rotations—effectively a manifestation of the Quantum Zeno effect—we can arbitrarily lower the probability of detonation while achieving perfect detection.

Define an infinitesimally small angle $\epsilon = \frac{\pi}{2n}$ for a very large integer n .

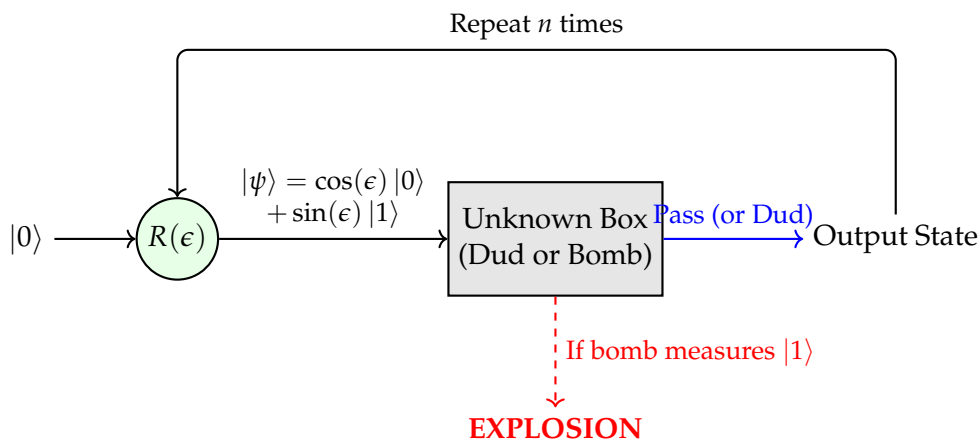


Figure 8: The iterative Elitzur-Vaidman bomb-testing protocol. The state is gently rotated away from $|0\rangle$, and the bomb (acting as a continuous measuring device) repeatedly collapses it back to $|0\rangle$, preventing the amplitude of $|1\rangle$ from accumulating.

The algorithm proceeds as follows:

1. Initialize a photon in state $|0\rangle$.
2. Apply a physical rotation device $R(\epsilon)$ to shift the state to $\cos(\epsilon)|0\rangle + \sin(\epsilon)|1\rangle$.
3. Inject the photon into the box.
4. If the photon survives, redirect it back to step 2. Repeat this loop strictly n times.
5. Perform a final measurement in the standard basis.

Case 1: The Dud. If the box is empty, it exerts no measurement. The photon undergoes n consecutive continuous rotations by ϵ . The final angle is exactly $n \times \frac{\pi}{2n} = \frac{\pi}{2} = 90^\circ$. The state deterministically rotates from $|0\rangle$ to $|1\rangle$. A standard measurement perfectly outputs 1.

Case 2: The Bomb. The bomb acts as a persistent measuring device in the standard basis. In each of the n iterations, the incoming state $\cos(\epsilon) |0\rangle + \sin(\epsilon) |1\rangle$ is measured. The probability of measuring $|1\rangle$ and detonating the bomb in a single iteration is:

$$P(\text{explode}) = \sin^2(\epsilon)$$

For small ϵ , the Taylor expansion of $\sin(\epsilon) \approx \epsilon$. Therefore, the probability of detonating in one pass is at most ϵ^2 . Crucially, if the photon survives, the bomb's measurement wave-function collapse forces the state exactly back to a pure $|0\rangle$. The rotation does not accumulate.

Over all n iterations, the upper bound of the total explosion probability is the union bound:

$$\begin{aligned} P(\text{Total Explosion}) &\leq n \cdot \sin^2(\epsilon) \\ &\approx n \cdot \epsilon^2 \\ &= n \left(\frac{\pi}{2n} \right)^2 \\ &= \frac{\pi^2}{4n} \end{aligned}$$

Because the explosion probability scales as $\mathcal{O}(1/n)$, we can arbitrarily suppress it by choosing a sufficiently large n (e.g., if $n = 1000$, $P \approx 0.25\%$). If we do not explode, the photon is collapsed back to $|0\rangle$ on the final pass. Measuring the output gives exactly 0.

By simply reading the final measurement—1 for a dud, 0 for a bomb—we decode the physical contents of the box with near-perfect reliability and negligible physical risk.

4.4 Quantum Mechanics Law 3: Unitary Transformations

Aside from destructive measurement, we require a mathematical framework governing how quantum states dynamically evolve or are computationally manipulated. The third axiom of quantum mechanics dictates that an isolated quantum state can undergo any linear transformation that rigorously preserves its vector length.

Because quantum states are probability distributions, the sum of their squared amplitude magnitudes must persistently equal 1. A matrix U acting on a complex vector space represents a valid physical quantum transformation if and only if it preserves this normalization.

Let $|\psi\rangle$ be an arbitrary state. The preservation of length requires:

$$\begin{aligned} \langle\psi|\psi\rangle &= (U|\psi\rangle)^\dagger (U|\psi\rangle) \\ \langle\psi|\psi\rangle &= \langle\psi| U^\dagger U |\psi\rangle \end{aligned}$$

For this to hold for all possible vectors $|\psi\rangle$, the transformation matrix must satisfy the fundamental definition of a *Unitary Matrix*:

$$U^\dagger U = I$$

where I is the identity matrix, and $U^\dagger$ represents the conjugate transpose (adjoint) of U . This structural requirement leads to several profound consequences for quantum computation:

1. **Orthonormal Columns:** The columns of any unitary matrix form a perfectly orthonormal basis. U essentially acts as a pure change-of-basis operator.
2. **Angle Preservation:** Unitary operations preserve not only lengths but the inner products (angles) between any pair of states. If $|\phi\rangle$ and $|\psi\rangle$ are transformed by U , their new inner product is $\langle\phi| U^\dagger U |\psi\rangle = \langle\phi|\psi\rangle$.
3. **Strict Reversibility:** Every unitary matrix is inherently invertible, and its inverse is exactly its conjugate transpose ($U^{-1} = U^\dagger$). Consequently, every operation in a quantum computer (excluding measurement) is perfectly, deterministically reversible. There is no information loss.

4.5 Fundamental Unitary Quantum Gates

The physical construction of quantum algorithms requires mapping computational "gates" to specific unitary matrices. The core single-qubit ($\mathbb{C}^2$) gates include:

Rotation Gate $R(\theta)$: Geometrically rotates the vector by an angle θ .

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Pauli-X Gate (Quantum NOT): Reflects the state across the 45° axis, interchanging amplitudes of $|0\rangle$ and $|1\rangle$.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hadamard Gate H : A reflection across the 22.5° axis ($\pi/8$), crucial for creating uniform superpositions from computational basis states. It also defines the discrete 2D Fourier transform.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Phase Gate S : Introduces a complex phase exclusively to the $|1\rangle$ component without altering geometric probabilities.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Pauli-Z Gate: Flips the sign of the $|1\rangle$ amplitude.

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Unitary transformations scale to higher dimensions perfectly. For two-qubit systems ($\mathbb{C}^4$), permutation matrices like the SWAP gate exchange the physical states of two particles without violating length constraints:

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

4.6 The Square Root of NOT ($\sqrt{X}$)

The continuous, linear algebra-driven nature of quantum mechanics permits transformations that have no classical analog. Consider the classical NOT gate, which deterministically swaps a 0 to a 1. In deterministic computing, there is no discrete operation that "half-NOTs" a bit such that applying it twice completes the flip.

However, since the quantum NOT gate (X) is a matrix, we can physically execute a transformation that acts as its perfect square root, $\sqrt{X}$, satisfying:

$$(\sqrt{X})^2 = X$$

Because the standard NOT gate acts as a reflection in real Euclidean space, it cannot be continuously rotated into a square root using solely real numbers. The instantiation of $\sqrt{X}$ strictly

mandates the use of complex amplitudes, manipulating the evolution through an imaginary spatial dimension:

$$\sqrt{X} = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$

We can rigorously verify its behavior through matrix multiplication:

$$\begin{aligned} (\sqrt{X})^2 &= \frac{1}{4} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} (1+i)^2 + (1-i)^2 & (1+i)(1-i) + (1-i)(1+i) \\ (1-i)(1+i) + (1+i)(1-i) & (1-i)^2 + (1+i)^2 \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} (2i) + (-2i) & 2+2 \\ 2+2 & (-2i) + (2i) \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} 0 & 4 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X \end{aligned}$$

5 Lecture 4.5: Discriminating Quantum States

5.1 The Fundamental Problem of Quantum State Discrimination

In our previous analysis of the Elitzur-Vaidman bomb tester, we implemented a protocol where a qubit is initialized and slightly rotated by an angle ϵ . Upon interacting with the experimental apparatus, if the box contains a "dud," the qubit simply passes through and remains at the angle ϵ . If the box contains a bomb and does not explode, the destructive measurement wave-function collapse forces the qubit into the state $|0\rangle$.

This scenario naturally prompts a profound theoretical question: since the emergent qubit is either in the state $|\epsilon\rangle$ (dud) or $|0\rangle$ (bomb), could we not simply perform a single measurement to tell the difference between these two states? Why do we strictly require repeating the rotation-measurement process $\mathcal{O}(1/\epsilon)$ times to drive the angle to 90° ?

This specific dilemma belongs to a foundational task in quantum information theory known as *quantum state discrimination*. Formally, suppose you are given an unknown quantum state $|\psi\rangle$ and a promise that $|\psi\rangle$ is identically equal to one of two known, non-orthogonal quantum states, $|u\rangle$ or $|v\rangle$. We assume the mathematical descriptions of $|u\rangle$ and $|v\rangle$ are fully known, and the geometric angle separating them in Hilbert space is θ .

The objective is to apply a sequence of unitary operations and measurements to the unknown state $|\psi\rangle$ to correctly guess whether $|\psi\rangle = |u\rangle$ or $|\psi\rangle = |v\rangle$. As we will demonstrate, this problem maps perfectly to the theoretical computer science classification of randomized algorithms: two-sided error, one-sided error, and zero-sided error.

5.2 Two-Sided Error Discrimination

In a two-sided error algorithm (analogous to the BPP complexity class in classical computing), the algorithm is permitted to make an incorrect guess regardless of whether the true state is $|u\rangle$ or $|v\rangle$.

Because unitary transformations preserve inner products (and consequently, geometric angles), we are free to rotate the coordinate system without altering the distinguishability of the states. To optimize our measurement, we apply a unitary transformation that rotates $|u\rangle$ and $|v\rangle$ such that their angle bisector aligns perfectly with the 45° diagonal ($\pi/4$) relative to the standard computational basis $\{|0\rangle, |1\rangle\}$.

Once the system is aligned, we perform a measurement in the standard basis $\{|0\rangle, |1\rangle\}$. The decision rule is straightforward:

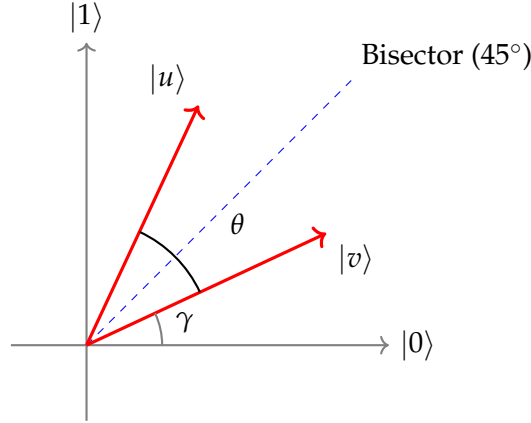


Figure 9: Geometric configuration for two-sided error discrimination. The states $|u\rangle$ and $|v\rangle$ are rotated such that their bisector lies at exactly 45° to the standard measurement basis.

- If the measurement yields 1, guess $|\psi\rangle = |u\rangle$.
- If the measurement yields 0, guess $|\psi\rangle = |v\rangle$.

Let us calculate the probability of error. Suppose the true state is indeed $|\psi\rangle = |u\rangle$. The algorithm produces an error if the readout is 0. The angle between $|u\rangle$ and $|0\rangle$ is $\gamma = 45^\circ + \frac{\theta}{2}$. According to the Born rule, the probability of measuring 0 is the squared magnitude of the amplitude:

$$\begin{aligned} P(\text{error} \mid |u\rangle) &= |\langle 0|u\rangle|^2 \\ &= \cos^2\left(45^\circ + \frac{\theta}{2}\right) \end{aligned}$$

Applying the trigonometric identity $\cos^2(x) = \frac{1}{2} + \frac{1}{2} \cos(2x)$, we obtain:

$$\begin{aligned} P(\text{error} \mid |u\rangle) &= \frac{1}{2} + \frac{1}{2} \cos(90^\circ + \theta) \\ &= \frac{1}{2} - \frac{1}{2} \sin(\theta) \end{aligned}$$

By symmetry, if the true state is $|v\rangle$, an error occurs when the readout is 1. The angle between $|v\rangle$ and $|1\rangle$ is identical ($45^\circ + \theta/2$), yielding the exact same error probability:

$$P(\text{error} \mid |v\rangle) = \frac{1}{2} - \frac{1}{2} \sin(\theta)$$

If $\theta \approx 0$ (the states are nearly identical), the error approaches 50%, equivalent to a random coin guess. If $\theta = 90^\circ$ (the states are orthogonal), the error collapses strictly to 0.

5.3 One-Sided Error Discrimination

A one-sided error algorithm (analogous to the RP or co-RP complexity classes) guarantees that under one of the conditions, it will absolutely never return a false positive.

To construct a zero false-positive algorithm for $|v\rangle$, we abandon the standard basis and instead design a physical measurement apparatus corresponding to the orthonormal basis $\{|u\rangle, |u^\perp\rangle\}$, where $|u^\perp\rangle$ is the state geometrically rotated 90° from $|u\rangle$.

The algorithm proceeds as follows:

- Measure the unknown state $|\psi\rangle$ in the $\{|u\rangle, |u^\perp\rangle\}$ basis.

- If the readout is $|u^\perp\rangle$, guess with absolute certainty that $|\psi\rangle = |v\rangle$.
- If the readout is $|u\rangle$, guess $|\psi\rangle = |u\rangle$.

Case 1: $|\psi\rangle = |u\rangle$. If the true state is $|u\rangle$, the probability of the device reading $|u^\perp\rangle$ is strictly $|\langle u^\perp | u \rangle|^2 = 0$. The device will output $|u\rangle$ with 100% probability, and we will correctly guess $|u\rangle$. The error in this case is exactly 0.

Case 2: $|\psi\rangle = |v\rangle$. If the true state is $|v\rangle$, we make an error if the device outputs $|u\rangle$ (leading us to guess $|u\rangle$). The probability of this occurring is the inner product squared between $|v\rangle$ and $|u\rangle$:

$$\begin{aligned} P(\text{error} \mid |v\rangle) &= |\langle u | v \rangle|^2 \\ &= \cos^2(\theta) \\ &= 1 - \sin^2(\theta) \end{aligned}$$

This framework elegantly explains the failure of the naive, single-shot Elitzur-Vaidman bomb test. If we measure a system separated by an infinitesimally small angle ϵ , the false-negative probability is $1 - \sin^2(\epsilon)$. As $\sin(\epsilon) \approx \epsilon$, this probability of failure is functionally 1. Attempting a one-sided discrimination on a slight angle yields essentially no usable information, mandating the repeated Zeno effect rotations to physically widen the angle θ to 90° .

5.4 Zero-Sided Error Discrimination (Unambiguous Discrimination)

The most stringent classical algorithmic paradigm is zero-sided error (a Las Vegas algorithm). In this model, the algorithm is never permitted to make a mathematically incorrect classification. If it cannot conclusively determine whether the state is $|u\rangle$ or $|v\rangle$, it must explicitly output a designated “I don’t know” failure state. The computational goal is to minimize the probability of this “don’t know” output.

We can naively construct a zero-sided error test by probabilistically combining two one-sided error tests. We have one test with no false positives for $|u\rangle$ (measuring in $\{|v\rangle, |v^\perp\rangle\}$), and one test with no false positives for $|v\rangle$ (measuring in $\{|u\rangle, |u^\perp\rangle\}$).

Naive Zero-Sided Algorithm:

1. Flip a fair classical coin.
2. If Heads (probability $1/2$), execute the no-false-positives-for- $|v\rangle$ test (measure in $\{|u\rangle, |u^\perp\rangle\}$). If the outcome is $|u^\perp\rangle$, output $|v\rangle$. If the outcome is $|u\rangle$, output “I don’t know”.
3. If Tails (probability $1/2$), execute the no-false-positives-for- $|u\rangle$ test (measure in $\{|v\rangle, |v^\perp\rangle\}$). If the outcome is $|v^\perp\rangle$, output $|u\rangle$. If the outcome is $|v\rangle$, output “I don’t know”.

Let us evaluate the probability of the algorithm returning “I don’t know”. Without loss of generality, assume the true state is $|v\rangle$. If the coin lands Heads (probability $1/2$), the algorithm performs the $\{|u\rangle, |u^\perp\rangle\}$ measurement. It outputs “I don’t know” if the measurement yields $|u\rangle$, which occurs with probability $\cos^2(\theta)$. If the coin lands Tails (probability $1/2$), the algorithm performs the $\{|v\rangle, |v^\perp\rangle\}$ measurement. Because the state is exactly $|v\rangle$, the measurement yields $|v\rangle$ with 100% probability, forcing an “I don’t know” output.

Thus, the total probability of an ambiguous “I don’t know” failure is:

$$\begin{aligned} P(\text{“Don’t Know”}) &= \frac{1}{2} \cos^2(\theta) + \frac{1}{2} (1) \\ &= \frac{1}{2} (1 - \sin^2(\theta)) + \frac{1}{2} \\ &= 1 - \frac{1}{2} \sin^2(\theta) \end{aligned}$$

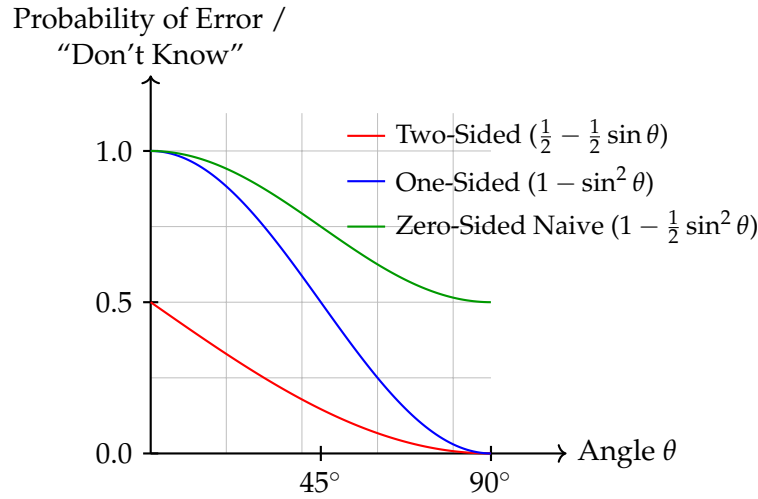


Figure 10: Comparative scaling of algorithmic failure probabilities as a function of the geometric separation θ between the two candidate states.

5.5 The Suboptimality of the Naive Approach and the Necessity of Ancillas

A critical failure of our naive zero-sided algorithm occurs at the asymptotic limit $\theta = 90^\circ$. If $|u\rangle$ and $|v\rangle$ are strictly orthogonal, they are perfectly distinguishable via a simple measurement in the $\{|u\rangle, |v\rangle\}$ basis. The probability of error should mathematically collapse to exactly 0.

However, evaluating our naive algorithm at $\theta = 90^\circ$ yields:

$$P(\text{"Don't Know"} \mid \theta = 90^\circ) = 1 - \frac{1}{2} \sin^2(90^\circ) = 1 - \frac{1}{2}(1) = \frac{1}{2}$$

This reveals that the coin-flip protocol is theoretically suboptimal. The fundamental limits of quantum mechanics dictate that the optimal success probability for zero-sided error discrimination is achieved when the “don’t know” outcome triggers with probability exactly $\cos(\theta)$. This converges properly to 0 as $\theta \rightarrow 90^\circ$.

Remarkably, this mathematically optimal $\cos(\theta)$ lower bound cannot be practically instantiated using exclusively single-qubit transformations and measurements. To implement the optimal unambiguous state discrimination protocol, one must expand the dimensionality of the state space. This requires initializing an auxiliary quantum particle (an *ancilla qubit*), executing a higher-dimensional unitary transformation on the joint multi-qubit system, and performing a composite measurement. The formalism required to describe such interacting multi-particle phenomena strictly dictates our transition into the linear algebra of tensor products and multi-qubit systems.

6 Lecture 5: Multi-Qubit Systems and Joint States

6.1 Higher-Dimensional Quantum Systems

While foundational quantum mechanics often introduces isolated two-level systems (qubits), physical reality frequently demands the analysis of higher-dimensional state spaces. A theoretical particle with spin-3/2, such as certain baryons, inherently possesses four distinct basic states, forming a dimension-four quantum system. The state of such a d -level system, generalized as a *qudit*, is governed by a unit vector of d complex amplitudes.

However, by far the most ubiquitous method of realizing a four-dimensional quantum state is not by discovering a single particle with four intrinsic levels, but rather by combining two

independent qubits. Consider a system composed of two spatially distinct photons, analyzing exclusively their polarization. If $|0\rangle$ represents horizontal polarization and $|1\rangle$ represents vertical polarization, the combined basis states of the two-photon system manifest as the four logical permutations: $|00\rangle$ (horizontal-horizontal), $|01\rangle$ (horizontal-vertical), $|10\rangle$ (vertical-horizontal), and $|11\rangle$ (vertical-vertical).

In full generality, the joint state vector $|\gamma\rangle$ of this composite system requires four complex amplitudes:

$$|\gamma\rangle = \gamma_{00}|00\rangle + \gamma_{01}|01\rangle + \gamma_{10}|10\rangle + \gamma_{11}|11\rangle$$

By quantum mechanical axioms, these amplitudes must satisfy the normalization constraint $\sum_{i,j \in \{0,1\}} |\gamma_{ij}|^2 = 1$. The identical scaling logic dictates that three coupled qubits reside in an 8-dimensional vector space (2^3), and 100 qubits—the threshold for classically intractable quantum computation—demand a state space of 2^{100} complex numbers.

6.2 Constructing Joint States from Independent Particles

Let us formalize the interaction of independent quantum systems. Suppose Alice possesses a qubit initialized in state $|\psi\rangle$ and Bob separately prepares a qubit in state $|\phi\rangle$.

$$\begin{aligned} |\psi\rangle &= \alpha_0|0\rangle + \alpha_1|1\rangle \\ |\phi\rangle &= \beta_0|0\rangle + \beta_1|1\rangle \end{aligned}$$

If they bring these spatially separated particles together into a shared physical apparatus without prior interaction, they form an “unentangled” joint system.

To deduce the amplitudes of the combined 4-dimensional state, we can draw a direct analogy to the mathematics of classical probabilistic computation. If Alice flips a biased coin with probability α_0 of landing heads (0), and Bob independently flips a coin with probability β_0 of landing heads (0), the joint probability of obtaining 00 is exactly the product $\alpha_0\beta_0$. The laws of quantum mechanics axiomatically impose the exact same algebraic structure for probability amplitudes:

$$\begin{aligned} \gamma_{00} &= \alpha_0\beta_0 \\ \gamma_{01} &= \alpha_0\beta_1 \\ \gamma_{10} &= \alpha_1\beta_0 \\ \gamma_{11} &= \alpha_1\beta_1 \end{aligned}$$

We must verify that this resulting composite vector remains a physically valid quantum state by checking its normalization. The sum of the squared magnitudes is:

$$\begin{aligned} \sum |\gamma_{ij}|^2 &= |\alpha_0\beta_0|^2 + |\alpha_0\beta_1|^2 + |\alpha_1\beta_0|^2 + |\alpha_1\beta_1|^2 \\ &= |\alpha_0|^2(|\beta_0|^2 + |\beta_1|^2) + |\alpha_1|^2(|\beta_0|^2 + |\beta_1|^2) \\ &= (|\alpha_0|^2 + |\alpha_1|^2)(|\beta_0|^2 + |\beta_1|^2) \end{aligned}$$

Since $|\psi\rangle$ and $|\phi\rangle$ are independently valid states, their respective squared amplitudes sum to 1. Therefore, the total sum is $1 \times 1 = 1$, strictly preserving unitarity.

6.3 The Tensor Product Formalism

In linear algebra, the mathematical operation that rigorously defines the merging of these vector spaces is the *tensor product* (or Kronecker product), denoted by the symbol $\otimes$.

For a d -dimensional qudit $|\psi\rangle$ and an e -dimensional qudit $|\phi\rangle$, their tensor product constructs a composite column vector of dimension $d \times e$. For our two-qubit example, it is explicitly defined as:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \otimes \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0 \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \\ \alpha_1 \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha_0\beta_0 \\ \alpha_0\beta_1 \\ \alpha_1\beta_0 \\ \alpha_1\beta_1 \end{pmatrix}$$

The tensor product formalism flawlessly reproduces our intuitive basis combinations. For instance:

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \equiv |00\rangle$$

Consider a more intricate example where Alice prepares her qubit in the basis state $|0\rangle$ and Bob prepares the uniform superposition state $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$. The tensor product yields:

$$|0\rangle \otimes |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle$$

Reversing the order fundamentally alters the physical state, demonstrating that the tensor product is inherently non-commutative:

$$|+\rangle \otimes |0\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

6.3.1 Matrix Tensor Products and Algebraic Properties

The Kronecker product gracefully extends to arbitrary matrices, allowing us to compute the tensor product of operators acting on multi-qubit systems. Let A be an $m \times n$ matrix and B be a $p \times q$ matrix. Their tensor product $A \otimes B$ is a massive block matrix of dimension $(mp) \times (nq)$:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ a_{21}B & a_{22}B & \dots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \dots & a_{mn}B \end{pmatrix}$$

The operator $\otimes$ behaves mechanically like a non-commutative multiplication and satisfies strict algebraic properties essential for quantum algorithm design:

1. **Distributivity:** $(A + B) \otimes C = (A \otimes C) + (B \otimes C)$
2. **Associativity:** $(A \otimes B) \otimes C = A \otimes (B \otimes C)$
3. **Conjugate Transpose:** $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ (Notice that the order of A and B does *not* reverse, unlike standard matrix multiplication $(AB)^\dagger = B^\dagger A^\dagger$).
4. **Mixed-Product Property:** $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$

6.4 Multi-Qubit Unitary Transformations: The CNOT Gate

Once combined, the joint 4-dimensional vector can undergo global unitary transformations. Synthesizing physical hardware capable of executing these multi-qubit interactions—such as precisely calibrated magnetic fields in trapped ion architectures—remains one of the paramount engineering challenges of quantum computing.

The canonical two-qubit gate is the **Controlled-NOT (CNOT)** gate. In classical logic, this is analogous to the reversible XOR gate. The CNOT gate designates one qubit as the “control” and the second as the “target”. Operationally:

- If the control qubit is $|0\rangle$, the target qubit remains unchanged.
- If the control qubit is $|1\rangle$, a quantum NOT gate (Pauli-X) is applied to the target qubit.

Formally, the CNOT maps the computational basis states as follows:

$$\begin{aligned} |00\rangle &\mapsto |00\rangle \\ |01\rangle &\mapsto |01\rangle \\ |10\rangle &\mapsto |11\rangle \\ |11\rangle &\mapsto |10\rangle \end{aligned}$$

Extracting the matrix representation directly from these basis mappings yields a 4×4 permutation matrix:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Since permutation matrices are strictly orthogonal (and real), CNOT is its own inverse: $\text{CNOT}^2 = I$.

6.5 Quantum Circuit Design: Constructing the EPR Pair

We can immediately apply these primitives to construct one of the most conceptually profound states in physics: the Bell state (or EPR pair, named after Einstein, Podolsky, and Rosen).

Consider the standard initialization of a two-qubit quantum computer into the pristine state $|00\rangle$. The generation of a Bell state requires exactly two logical steps: 1. Apply a single-qubit Hadamard (H) gate to the first qubit. 2. Apply a joint CNOT gate, using the first qubit as control and the second as target.

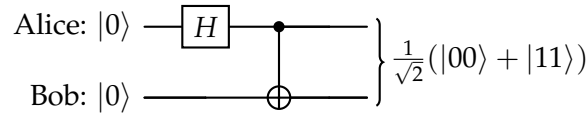


Figure 11: Quantum circuit diagram for the generation of the Bell state (EPR pair). The Hadamard gate places the control qubit into a superposition, and the subsequent CNOT entangles the joint system.

Let us rigorously step through the temporal evolution of the joint state.

Time 1 (Initialization):

$$|\psi_1\rangle = |0\rangle \otimes |0\rangle = |00\rangle$$

Time 2 (Post-Hadamard): Alice applies H to her qubit, mapping $|0\rangle \rightarrow |+\rangle$. Bob's qubit is completely unaffected.

$$\begin{aligned} |\psi_2\rangle &= (H|0\rangle) \otimes |0\rangle = |+\rangle \otimes |0\rangle \\ &= \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) \otimes |0\rangle \\ &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle \end{aligned}$$

Time 3 (Post-CNOT): The joint state passes through the 4×4 CNOT matrix. By linearity, the CNOT acts independently on the superposition terms. The $|00\rangle$ term satisfies the control=0 condition, remaining unchanged. The $|10\rangle$ term satisfies the control=1 condition, causing the target qubit to flip to $|11\rangle$.

$$\begin{aligned} |\psi_3\rangle &= \text{CNOT} \left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle \right) \\ &= \frac{1}{\sqrt{2}}\text{CNOT}|00\rangle + \frac{1}{\sqrt{2}}\text{CNOT}|10\rangle \\ &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \end{aligned}$$

This resulting vector is the foundational Bell State.

6.6 Formal Definition of Entanglement

The mathematical architecture of the Bell state triggers a collapse of classical physical intuitions, prompting the formal definition of quantum entanglement.

Theorem: *The Bell state cannot be expressed as the tensor product of any two independent single-qubit states.*

Proof: Assume for the sake of contradiction that there exist two valid qubit states $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|\phi\rangle = \beta_0|0\rangle + \beta_1|1\rangle$ such that their joint unentangled state perfectly equals the Bell state:

$$\begin{aligned} |\psi\rangle \otimes |\phi\rangle &= \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle \\ &\equiv \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + 0|10\rangle + \frac{1}{\sqrt{2}}|11\rangle \end{aligned}$$

For any generic tensor product, notice the strict algebraic relationship governing the amplitudes. The product of the outer amplitudes intrinsically equals the product of the inner amplitudes:

$$\begin{aligned} (\text{Amplitude}_{00}) \times (\text{Amplitude}_{11}) &= (\alpha_0\beta_0)(\alpha_1\beta_1) = \alpha_0\alpha_1\beta_0\beta_1 \\ (\text{Amplitude}_{01}) \times (\text{Amplitude}_{10}) &= (\alpha_0\beta_1)(\alpha_1\beta_0) = \alpha_0\alpha_1\beta_0\beta_1 \end{aligned}$$

However, evaluating this invariant on the target Bell state yields:

$$\begin{aligned} (\text{Amplitude}_{00}) \times (\text{Amplitude}_{11}) &= \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{2} \\ (\text{Amplitude}_{01}) \times (\text{Amplitude}_{10}) &= (0)(0) = 0 \end{aligned}$$

Since $1/2 \neq 0$, the initial assumption forces a mathematical contradiction. The proof concludes. ■

We categorically define any composite quantum state that cannot be factored into the tensor product of individual subsystems as **entangled**. Entangled states embed non-local correlations that do not exist classically; determining the state of one partition mathematically forces instantaneous knowledge regarding the opposite partition, entirely bypassing the limitations of classical independence. Conversely, states structurally identical to $|\psi\rangle \otimes |\phi\rangle$ are defined as **unentangled**.

6.7 Local Unitary Transformations on Joint Spaces

Once particles are entangled, tracking their physical evolution strictly mandates treating them as a monolithic 4-dimensional vector. Nonetheless, reality physically separates these entities. If Alice physically distances her qubit across a university campus, she retains the capacity to apply localized unitary operations exclusively to her isolated sub-system.

We must reconcile the application of a local 2×2 unitary transformation U with the requirement of a global 4×4 state update. The tensor product provides the requisite mathematical interface. If Bob applies an arbitrary unitary $U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ to the second qubit while Alice applies no operation to the first, Alice effectively applies the 2×2 Identity matrix I . The resultant global transformation is inherently represented by $I \otimes U$.

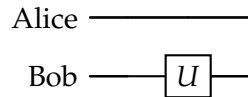


Figure 12: A local unitary applied to the target qubit of a joint system strictly maps to the 4×4 global unitary operator $I \otimes U$.

Expanding this block matrix using the Kronecker definitions yields the exact 4×4 unitary necessary to compute the updated joint state:

$$\begin{aligned} I \otimes U &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes U \\ &= \begin{pmatrix} 1 \cdot U & 0 \cdot U \\ 0 \cdot U & 1 \cdot U \end{pmatrix} \\ &= \begin{pmatrix} p & q & 0 & 0 \\ r & s & 0 & 0 \\ 0 & 0 & p & q \\ 0 & 0 & r & s \end{pmatrix} \end{aligned}$$

If we apply this matrix to the basis state $|00\rangle$, it accurately maps exclusively the target qubit's subspace while leaving the control rigid:

$$(I \otimes U) |00\rangle = \begin{pmatrix} p & q & 0 & 0 \\ r & s & 0 & 0 \\ 0 & 0 & p & q \\ 0 & 0 & r & s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ r \\ 0 \\ 0 \end{pmatrix} \equiv p |00\rangle + r |01\rangle$$

This elegantly coincides with the independent tensor evaluation $|0\rangle \otimes (U|0\rangle)$, preserving the rigor of the linear algebraic formalism alongside the localized physical mechanics. Any generic combination of simultaneous local operations U on Alice's qubit and V on Bob's qubit is universally resolved via the joint operator $U \otimes V$.

7 Lecture 5.5: Global Phase and the Indistinguishability of Quantum States

7.1 Symmetry in State Discrimination Beyond Orthogonality

In the formal analysis of quantum state discrimination, we previously established the theoretical limits of distinguishing between two known, non-orthogonal quantum states, $|u\rangle$ and $|v\rangle$, separated by a geometric angle θ . We observed how the minimal probability of error varies

depending on whether the algorithmic protocol permits two-sided error, one-sided error, or zero-sided error.

Previously, our analysis constrained the angle θ to the interval $[0^\circ, 90^\circ]$ —ranging from completely identical states ($\theta = 0^\circ$) to strictly orthogonal states ($\theta = 90^\circ$). A natural extension in linear algebra dictates that vectors in a plane can subtend obtuse angles up to 180° . We must now rigorously analyze the discriminatory bounds for $\theta \in (90^\circ, 180^\circ]$.

Geometrically, distinguishing between a target state $|v\rangle$ and an obtuse state $|u\rangle$ is structurally identical to distinguishing between $|v\rangle$ and the antipodal state $-|u\rangle$. As illustrated in the coordinate space, flipping $|u\rangle$ to $-|u\rangle$ maps the separating angle from θ to exactly $180^\circ - \theta$.

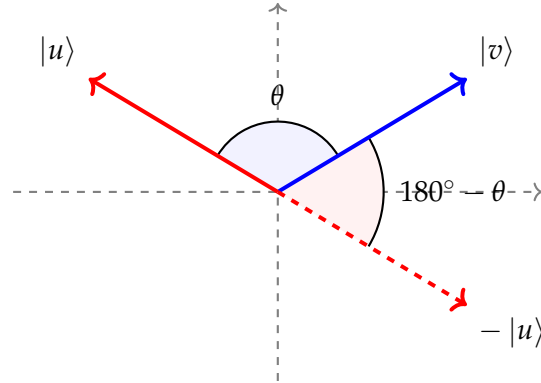


Figure 13: Symmetry of quantum state discrimination. The problem of differentiating between $|v\rangle$ and $|u\rangle$ (separated by an obtuse angle θ) is computationally and physically equivalent to distinguishing between $|v\rangle$ and $-|u\rangle$ (separated by the acute angle $180^\circ - \theta$).

Due to this absolute symmetry, the algorithmic error curves evaluated from 0° to 90° mirror themselves perfectly across the 90° axis, extending to 180° . At $\theta = 180^\circ$, the capacity to distinguish the states degenerates completely, returning to the maximum error threshold of 50%. This means any attempt to distinguish between the parallel but oppositely directed states $|u\rangle$ and $-|u\rangle$ yields no information greater than a uniformly random coin guess.

7.2 The Absolute Indistinguishability of $|u\rangle$ and $-|u\rangle$

The mathematical collapse of discriminatory power at 180° dictates a profound physical reality: the quantum state $|u\rangle$ and the quantum state $-|u\rangle$ are completely physically indistinguishable.

Let us formalize this impossibility. Suppose an oracle provides an unknown state $|\psi\rangle$ with the strict promise that $|\psi\rangle \in \{|u\rangle, -|u\rangle\}$. The only physically permissible operations on $|\psi\rangle$ are unitary transformations and measurements.

1. Unitary Transformations: Applying any arbitrary unitary operator R to the states preserves their geometry. If we rotate the system such that $|u\rangle$ aligns with the computational basis vector $|0\rangle$, the antipodal vector $-|u\rangle$ deterministically maps to $-|0\rangle$.

$$\begin{aligned} R|u\rangle &= |0\rangle \\ R(-|u\rangle) &= -R|u\rangle = -|0\rangle \end{aligned}$$

Rotations alter the coordinate representation but do not change the underlying physical relationship: the vectors remain perfectly anti-aligned.

2. Measurement Outcomes: Suppose we subject the rotated state to a standard computational basis measurement, probing for $|0\rangle$. According to the Born Rule, the probability of the measurement apparatus outputting 0 depends exclusively on the squared magnitude of the

complex amplitude.

$$\begin{aligned} P(0 \mid |0\rangle) &= |\langle 0|0\rangle|^2 = |1|^2 = 1 \\ P(0 \mid -|0\rangle) &= |\langle 0|-|0\rangle|^2 = |-1|^2 = 1 \end{aligned}$$

Both states trigger the 0 readout with 100% probability.

3. Post-Measurement State Collapse: Following measurement, the wavefunction irreversibly collapses onto the observed eigenstate. In both cases, the readout is 0. Therefore, if the initial state was $|0\rangle$, it collapses to $|0\rangle$. If the initial state was $-|0\rangle$, it mathematically collapses to $-|0\rangle$ (which, again, will yield identical probabilities for all future measurements). The measurement strictly yields zero classical bits of mutual information regarding the initial negative sign.

7.3 The Global Phase Factor

This indistinguishability extends far beyond the negative scalar -1 . For any complex number $c \in \mathbb{C}$ such that the magnitude $|c| = 1$, the state $|u\rangle$ and the state $c|u\rangle$ are physically identical.

Consider a generic measurement in an arbitrary basis $\{|w_i\rangle\}$. The probability of measuring outcome i for the state $c|u\rangle$ is:

$$\begin{aligned} P(i) &= |\langle w_i|cu\rangle|^2 \\ &= |c \cdot \langle w_i|u\rangle|^2 \\ &= |c|^2 \cdot |\langle w_i|u\rangle|^2 \end{aligned}$$

Since $|c| = 1$, this simplifies exactly to:

$$P(i) = 1 \cdot |\langle w_i|u\rangle|^2 = |\langle w_i|u\rangle|^2$$

The probability distributions across all conceivable measurement bases are mathematically equivalent.

The scalar coefficient c is formally defined as a **global phase**. A global phase can take the form of $c = -1$, $c = i$, or the generalized Euler representation $c = e^{i\phi}$ for any real angle ϕ . While local relative phases within a superposition (e.g., the difference between $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$) drastically alter interference patterns and measurement statistics, multiplying the entire state vector by a global phase induces absolutely no physically measurable difference.

This introduces a degree of “clunkiness” or representational redundancy into Dirac notation: distinct mathematical objects (like $|0\rangle$ and $-|0\rangle$) refer to the exact same ontological object in the physical universe. This mathematical redundancy will be formally resolved in subsequent lectures through the introduction of *density matrices* (mixed states), which intrinsically factor out global phases.

7.4 Prelude to Mixed States: Indistinguishable Ensembles

To preview the complexity of state representation, let us consider the indistinguishability of macroscopic probability distributions, known as **mixed states** or statistical ensembles.

Suppose an adversarial friend prepares a quantum particle using one of two distinct classical randomized protocols, denoted as Scenario ρ_1 and Scenario ρ_2 .

- **Scenario ρ_1 :** The friend flips a fair classical coin. If Heads, they initialize the qubit strictly in the state $|0\rangle$. If Tails, they initialize it strictly in the state $|1\rangle$.
- **Scenario ρ_2 :** The friend flips a fair classical coin. If Heads, they initialize the qubit strictly in the diagonal state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. If Tails, they initialize it in the anti-diagonal state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

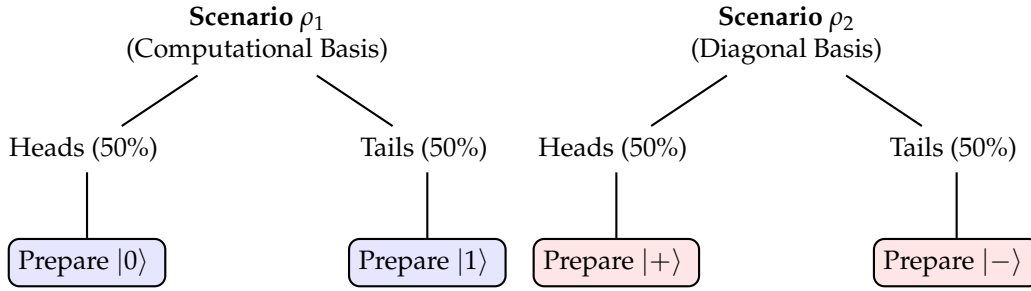


Figure 14: Two distinct classical randomized procedures for preparing a quantum state. Despite the radically different underlying quantum bases utilized during preparation, the resulting mixed states are perfectly indistinguishable to any observer.

Once the particle is prepared according to one of these stochastic processes, the coin is hidden, and the single unmeasured qubit $|\psi\rangle$ is handed to you. Your task is to guess whether the particle was generated via Scenario ρ_1 or Scenario ρ_2 .

Remarkably, despite the fact that the underlying state vectors drawn in these ensembles are fundamentally distinct, there exists absolutely no physical measurement, sequence of unitary transformations, or algorithmic protocol that can distinguish between these two scenarios with a probability better than random guessing (50%).

This phenomenon demonstrates that classical ignorance (probabilistic mixing) combined with quantum superposition creates entirely new classes of indistinguishable states. A rigorous mathematical formalism capable of resolving these equivalences and handling multi-qubit probabilistic distributions will be the core subject of the density matrix formalisms in upcoming discussions.

8 Lecture 6: Partial Measurements and Quantum Non-Locality

8.1 Independent Unitary Operations on Composite Systems

To establish the formal framework for partial measurements, we must first rigorously analyze the evolution of independent subsystems under localized unitary transformations. Suppose two separate parties, Alice and Bob, prepare isolated single-qubit states, $|\psi\rangle$ and $|\phi\rangle$ respectively. If these particles physically interact or are jointly prepared, they form a composite two-qubit state described by the tensor product space $\mathbb{C}^2 \otimes \mathbb{C}^2$. The generic unentangled joint state is defined as:

$$|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle$$

Once entangled or jointly constructed, Alice and Bob may separate. If Alice applies a local 2×2 unitary transformation U strictly to her particle, while Bob applies a 2×2 unitary V to his, the global operation is mathematically represented by the tensor product of the operators, $U \otimes V$.

Crucially, the linear algebraic properties of the Kronecker product guarantee that independent operations on distinct subsystems strictly commute. Applying Alice's gate followed by Bob's gate yields the exact same state evolution as applying them simultaneously or in reverse order:

$$(I \otimes V)(U \otimes I)|\Psi\rangle = (U \otimes V)|\Psi\rangle = (U \otimes I)(I \otimes V)|\Psi\rangle$$

This commutativity extends fundamentally to quantum measurements. If spatial separation ensures that localized operations act strictly on independent subsystems, the order in which Alice and Bob measure their respective particles does not alter the underlying statistical probabilities nor the final collapsed state.

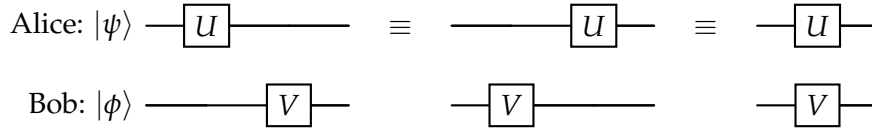


Figure 15: Commutativity of local operations. Sequential local unitary transformations on separate qubits are algebraically and physically equivalent to simultaneous execution.

8.2 The Mathematics of Partial Measurements

Consider a generic, potentially entangled two-qubit state $|\Psi\rangle$ shared between Alice and Bob:

$$|\Psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

where the complex amplitudes satisfy the normalization condition $\sum_{i,j \in \{0,1\}} |\alpha_{ij}|^2 = 1$.

Suppose Alice performs a partial measurement strictly on the first qubit using the standard computational basis $\{|0\rangle, |1\rangle\}$. The physical readout and the subsequent state collapse are governed by the following axioms:

1. Probability of Outcomes: The probability that Alice's measuring device outputs 0, denoted $P(0)$, is strictly the sum of the squared magnitudes of the amplitudes corresponding to basis states where Alice's qubit is $|0\rangle$:

$$P(0) = |\alpha_{00}|^2 + |\alpha_{01}|^2$$

Correspondingly, the probability of reading 1 is:

$$P(1) = |\alpha_{10}|^2 + |\alpha_{11}|^2$$

2. Partial Wave-Function Collapse: If Alice measures 0, the joint state vector irreversibly collapses. We systematically annihilate all components inconsistent with the readout (i.e., those beginning with $|1\rangle$) and renormalize the surviving components by dividing by the square root of the outcome's probability $\sqrt{P(0)}$. The post-measurement state $|\Psi'\rangle$ becomes:

$$\begin{aligned} |\Psi'\rangle &= \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{P(0)}} \\ &= |0\rangle \otimes \left(\frac{\alpha_{00}|0\rangle + \alpha_{01}|1\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \right) \end{aligned}$$

A profound consequence of this collapse is that the resulting post-measurement state is strictly *unentangled*. Factoring out Alice's $|0\rangle$ tensor component isolates Bob's qubit into a well-defined, pure single-qubit state.

Remark on Global Phase: The normalization factor must be the strictly positive real number $\sqrt{P(0)}$. Even if the isolated amplitudes (e.g., α_{00}) are complex or negative, multiplying a quantum state by a global phase factor (such as -1 or $e^{i\theta}$) produces a physically indistinguishable state. There is no physical experiment capable of differentiating between $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$.

8.3 Pure vs. Mixed States

If Alice takes her qubit and departs for the moon without communicating her measurement outcome to Bob, how should Bob mathematically model his isolated particle?

Since Bob does not know whether Alice's measurement yielded 0 or 1, his particle cannot be described by a single state vector (a *pure state*). Instead, the mathematically rigorous description is a classical probability distribution over pure quantum states, known formally as a **mixed state**.

Bob's mixed state is defined as:

- With probability $P(0)$, his state is $|\phi_0\rangle = \frac{\alpha_{00}|0\rangle + \alpha_{01}|1\rangle}{\sqrt{P(0)}}$.
- With probability $P(1)$, his state is $|\phi_1\rangle = \frac{\alpha_{10}|0\rangle + \alpha_{11}|1\rangle}{\sqrt{P(1)}}$.

To avoid the mathematically cumbersome task of tracking classical probability distributions throughout algorithmic evolutions, quantum computing protocols overwhelmingly defer intermediate measurements to the absolute end of the circuit. This ensures that the system globally evolves as a pure state under strict unitary operations until readout.

8.4 Einstein, EPR, and Spooky Action at a Distance

The principles of partial measurement generate profound philosophical and physical anomalies when applied to maximally entangled systems. Consider the canonical Einstein-Podolsky-Rosen (EPR) pair, also known as the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Contemporary quantum engineering routinely prepares macroscopic separations of these states; implementations have verified entanglement across distances of 143 kilometers in the Canary Islands, and up to 1,400 kilometers between ground stations and the *Micius* satellite in orbit.

Suppose Alice travels to the moon with the first qubit while Bob remains on Earth with the second. If Alice measures her qubit in the standard basis:

- With 50% probability, she measures 0. The global state instantaneously collapses to $|00\rangle$. Bob's state is definitively $|0\rangle$.
- With 50% probability, she measures 1. The global state instantaneously collapses to $|11\rangle$. Bob's state is definitively $|1\rangle$.

Immediately upon reading her instrument, Alice possesses absolute knowledge of Bob's physical state, despite the spacelike separation. Albert Einstein famously derided this instantaneous correlation as *spukhafte Fernwirkung* ("spooky action at a distance"). It seemingly violates the relativistic bound on information propagation.

Einstein conjectured that these correlations implied a **local hidden variable theory**. Under this hypothesis, the photons possess a pre-determined, locally stored classical state (like two synchronized classical coins flipped on Earth, covered, and then separated). The 50/50 probability would merely reflect experimental ignorance, not an underlying physical superposition.

However, John Bell later proved (and subsequent CHSH experiments empirically confirmed) that no local hidden variable theory can reproduce the statistical predictions of quantum mechanics. The true state of the particles prior to measurement is strictly the holistic superposition $|\Phi^+\rangle$.

8.5 Basis Independence and the No-Communication Theorem

The "spookiness" deepens when Alice recognizes she possesses the volition to measure her qubit in an alternative basis. Suppose she chooses to measure in the diagonal basis $\{|+\rangle, |-\rangle\}$, where:

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \\ |-\rangle &= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \end{aligned}$$

To analyze the partial measurement in this basis, we substitute the inverse relations ($|0\rangle = \frac{|+\rangle+|-\rangle}{\sqrt{2}}$ and $|1\rangle = \frac{|+\rangle-|-\rangle}{\sqrt{2}}$) into Alice's portion of the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \right) \otimes |0\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} |+\rangle - \frac{1}{\sqrt{2}} |-\rangle \right) \otimes |1\rangle$$

Applying the distributive property of the tensor product:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{2} |++\rangle + \frac{1}{2} |+-\rangle + \frac{1}{2} |-+\rangle - \frac{1}{2} |--\rangle \\ &= |+\rangle \otimes \left(\frac{1}{2} |0\rangle + \frac{1}{2} |1\rangle \right) + |-\rangle \otimes \left(\frac{1}{2} |0\rangle - \frac{1}{2} |1\rangle \right) \end{aligned}$$

Factoring the terms reveals a remarkable symmetry:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} |++\rangle + \frac{1}{\sqrt{2}} |--\rangle$$

Algebraically, the Bell state is structurally identical whether written in the standard basis or the diagonal basis.

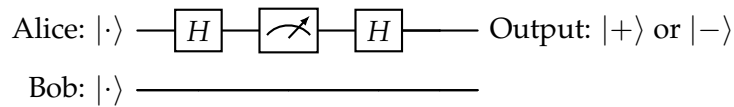


Figure 16: Simulating a partial measurement in the diagonal $\{|+\rangle, |-\rangle\}$ basis. Alice rotates the basis via a Hadamard gate, performs a standard computational measurement, and reverses the rotation (since $H = H^{-1}$).

If Alice measures in the $\{|+\rangle, |-\rangle\}$ basis, she obtains $+$ with 50% probability (collapsing Bob's state to $|+\rangle$) and $-$ with 50% probability (collapsing Bob's state to $|-\rangle$).

It appears Alice has discovered a mechanism to instantaneously transmit classical bits. By choosing her measurement basis, she can remotely force Bob's particle into one of two distinct mixed states:

- **Scenario ρ_1 (Alice measures Standard):** Bob's state is 50% $|0\rangle$ and 50% $|1\rangle$.
- **Scenario ρ_2 (Alice measures Diagonal):** Bob's state is 50% $|+\rangle$ and 50% $|-\rangle$.

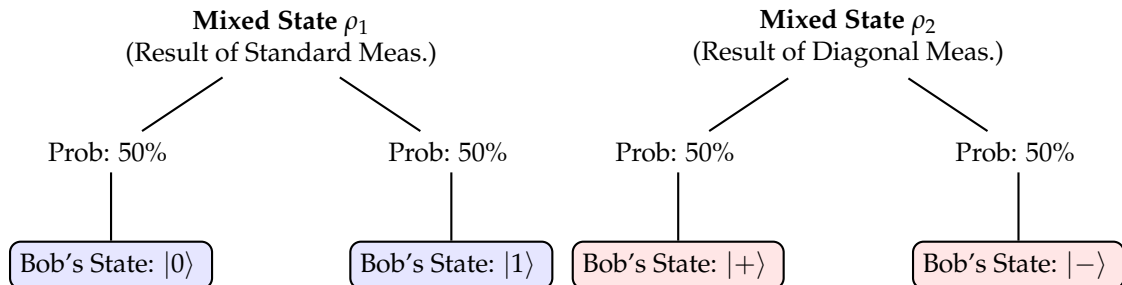


Figure 17: The two apparent mixed state ensembles for Bob based on Alice's choice of measurement basis. Despite the distinct pure states composing each ensemble, they are mathematically identical in terms of observational statistics.

The profound resolution to this paradox lies in the indistinguishability of mixed states. There exists no unitary sequence or quantum measurement Bob can perform to tell whether he holds

the distribution ρ_1 or ρ_2 . If Bob measures ρ_1 in the standard basis, he sees 0 (50%) and 1 (50%). If he measures ρ_2 in the standard basis, the inner product amplitudes ($|\langle 0|+\rangle|^2 = 0.5$) guarantee he still sees 0 (50%) and 1 (50%).

Because ρ_1 and ρ_2 generate absolutely identical probability distributions across all conceivable observational bases, Bob cannot decode Alice's choice. Therefore, while entanglement establishes non-local correlations, the **No-Communication Theorem** strictly dictates that these correlations cannot be weaponized to transmit information faster than the speed of light.

9 Lecture 7: The CHSH Game and Quantum Non-Locality

9.1 Recap: Partial Measurements and Spooky Action

In our previous discussion regarding spatially separated qubits, we explored the phenomenon that Albert Einstein notoriously dubbed *spukhafte Fernwirkung*, or “spooky action at a distance.” Let us formalize the setting: Alice and Bob prepare a pair of photons in the canonical Einstein-Podolsky-Rosen (EPR) state, also known as the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Following the initialization, Alice takes her photon to Mars, and Bob takes his to Jupiter. Despite the immense spatial separation, the particles remain perfectly entangled.

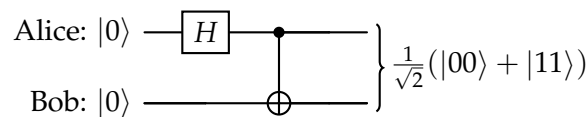


Figure 18: Quantum circuit for the preparation of the EPR pair ($|\Phi^+\rangle$). A Hadamard gate establishes superposition, followed by a CNOT gate to maximally entangle the spatial modes.

If Alice measures her qubit in the standard computational basis (which we shall refer to as the **Red basis**), she obtains the outcome 0 or 1 with equal probability. Instantaneously, the global wave-function collapses, and Bob's state strictly snaps to exactly the same state Alice measured ($|0\rangle$ or $|1\rangle$).

Crucially, we demonstrated that the EPR state possesses a profound symmetry. We can perfectly rewrite the state tensor in the diagonal basis ($\{|+\rangle, |-\rangle\}$), which we shall call the **Yellow basis**:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|++\rangle + \frac{1}{\sqrt{2}}|--\rangle$$

If Alice volitionally chooses to measure her qubit in the Yellow basis, she obtains $+$ or $-$ with a 50% probability each. Consequently, Bob's qubit collapses immediately to the corresponding $|+\rangle$ or $|-\rangle$ vector. While this capacity for Alice to remotely dictate the measurement basis of Bob's physical particle appears to allow for faster-than-light communication, the indistinguishability of mixed states ensures that, from Bob's local perspective, his density matrix remains identically random. Thus, the No-Communication Theorem holds securely.

However, the question remains: Can this non-local correlation be harnessed to perform a task strictly impossible under the classical laws of physics? This lecture answers that question in the affirmative through the *CHSH Game*.

9.2 The Setup of the CHSH Game

The CHSH game (named after John Clauser, Michael Horne, Abner Shimony, and Richard Holt, who adapted John Bell's 1964 theorem into a physical experiment in 1969) operates as a cooperative "magic trick" between Alice and Bob, mediated by neutral referees.

The geographical setup ensures strict spatial separation. Alice and her referee reside on Mars, while Bob and his referee reside on Jupiter. This guarantees that light cannot travel between them during the duration of a single experimental trial, strictly prohibiting any form of classical communication.

The protocol proceeds as follows:

1. **The Challenges:** At an agreed-upon, synchronized time, the referee on Mars flips a fair classical coin and issues Alice a challenge: either the **Red** basis or the **Yellow** basis. Simultaneously, the referee on Jupiter flips an independent fair coin and issues Bob a challenge: either the **Orange** basis or the **Green** basis.
2. **The Responses:** Within an extremely tight time window (e.g., 10 seconds), Alice and Bob must each output a single classical bit: 0 or 1.

To understand the objective, we must formally define the geometric relationship between these four conceptual bases. In the 2D plane of real amplitudes, we map the measurement bases at consecutive rotational intervals of $\frac{\pi}{8}$ radians (22.5°).

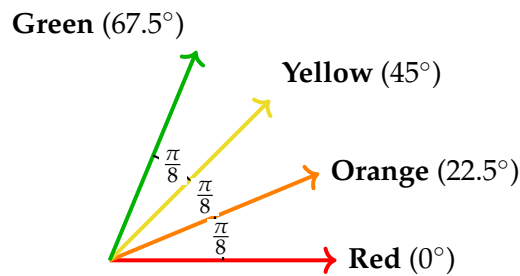


Figure 19: Geometric configuration of the four measurement bases. Each successive basis is separated by an angle of 22.5° ($\frac{\pi}{8}$). Notice that Red and Green are separated by an extreme angle of 67.5° , pushing them far apart.

The success condition of the game is determined by a bipartite matching graph. Alice and Bob win the game if they satisfy the following logical constraints based on the challenge pairs they receive:

- If (Red, Orange) $\rightarrow$ They must output the **Same** bit (00 or 11).
- If (Yellow, Orange) $\rightarrow$ They must output the **Same** bit (00 or 11).
- If (Yellow, Green) $\rightarrow$ They must output the **Same** bit (00 or 11).
- If (Red, Green) $\rightarrow$ They must output the **Opposite** bit (01 or 10).

9.3 Theoretical Bounds on Classical Strategies

Let us evaluate the theoretical limit of success for Alice and Bob assuming they are bound entirely by classical physics and local realism. We model their potential strategies in an ascending hierarchy of computational power.

1. Deterministic Strategy: Assume Alice and Bob operate as purely deterministic agents. Their strategy reduces to assigning a rigid output function mapping their local challenge to a

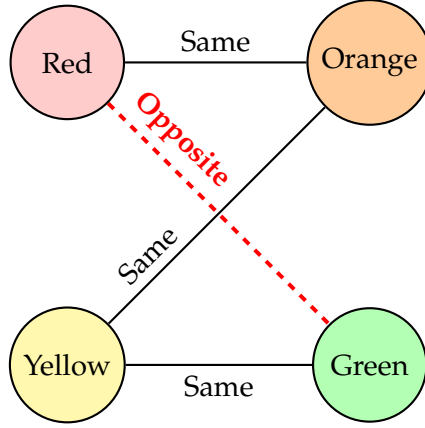


Figure 20: The Bipartite Challenge Graph defining the success conditions for the CHSH game. Alice receives nodes on the left; Bob receives nodes on the right. Because the cycle demands three equivalences and one negation, it contains a topological frustration.

binary output $\{0, 1\}$. Effectively, this constitutes assigning a static 0 or 1 label to each of the four nodes in our bipartite graph. Let $A(R), A(Y) \in \{0, 1\}$ and $B(O), B(G) \in \{0, 1\}$. The game demands:

$$\begin{aligned} A(R) &= B(O) \\ A(Y) &= B(O) \\ A(Y) &= B(G) \\ A(R) &\neq B(G) \end{aligned}$$

By transitivity, the first three equations force $A(R) = A(Y) = B(O) = B(G)$. However, the final constraint strictly demands $A(R) \neq B(G)$, generating an immediate algebraic contradiction. It is mathematically impossible to assign bits to the four vertices such that all four edge conditions are satisfied. The best possible deterministic assignment satisfies exactly 3 out of the 4 edges. Since the referees pick the challenge pair uniformly at random, the maximum deterministic success probability is exactly:

$$P_{\text{success}}^{\text{det}} = \frac{3}{4} = 75\%$$

2. Private Randomness: To be charitable, we permit Alice and Bob to operate as randomized agents (the BPP complexity model). Upon receiving a challenge, Alice may flip an arbitrary number of private local coins to determine her output. Because the expected value of any probabilistic mixture of deterministic strategies can never exceed the maximum of those individual deterministic strategies, the upper bound fundamentally remains:

$$P_{\text{success}}^{\text{private}} \leq 75\%$$

3. Shared Randomness (Local Hidden Variables): The most robust classical formulation permits Alice and Bob to share arbitrarily large strings of correlated random bits pre-generated on Earth before they depart. This setup perfectly formalizes Einstein's **Local Hidden Variables** hypothesis. The shared random string λ acts as the hidden local state determining the behavior of the particles. Even in this scenario, once the shared random variable λ is fixed, Alice and Bob's behavior conditionally reduces back to a deterministic mapping. For any specific coin flip sequence $\lambda = \lambda_0 \lambda_1 \dots$, the conditional success probability is bound by 75%. Integrating over all possible shared coin flip sequences yields:

$$P_{\text{success}}^{\text{classical}} = \sum_{\lambda} P(\lambda) \cdot P_{\text{success}}(\text{strategy} \mid \lambda) \leq 75\%$$

This absolute bound of 75% constitutes the mathematical core of **Bell's Inequality**. Any physical theory strictly obeying local realism cannot exceed this threshold.

9.4 The Quantum Strategy: Breaking Bell's Inequality

We now grant Alice and Bob the power of quantum mechanics. They pre-share an EPR pair $|\Phi^+\rangle$ on Earth and separate to their respective planets. Their strategy is elegantly simple:

- **Alice:** Modulates her measurement apparatus to match the angle of her requested challenge basis (Red at 0° or Yellow at 45°). She measures her half of the EPR pair and outputs the resulting bit directly.
- **Bob:** Modulates his measurement apparatus to match the angle of his requested challenge basis (Orange at 22.5° or Green at 67.5°). He measures his half of the EPR pair and outputs the resulting bit directly.

Since local operations on separated subsystems commute, we can analyze the sequence as if Alice measures first. Suppose the challenge is (Red, Orange). Alice measures in the Red basis. Her qubit collapses to a standard vector, and Bob's identical particle simultaneously collapses into the exact same vector. Now Bob is holding Alice's Red vector, but he must measure it in the Orange basis. Because the Orange basis is rotated precisely $\frac{\pi}{8} = 22.5^\circ$ relative to the Red basis, Born's Rule dictates the probability of Bob obtaining the *same* outcome (the projection onto the nearest basis vector) is the cosine squared of the angle:

$$P(\text{Same}) = \cos^2(22.5^\circ) = \cos^2\left(\frac{\pi}{8}\right)$$

Applying the half-angle trigonometric identity $\cos^2(\theta) = \frac{1}{2} + \frac{1}{2}\cos(2\theta)$:

$$\begin{aligned}\cos^2\left(\frac{\pi}{8}\right) &= \frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi}{4}\right) \\ &= \frac{1}{2} + \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right) \\ &\approx 0.853 \quad (85.3\%)\end{aligned}$$

This success rate of $\approx 85\%$ holds symmetrically for the (Yellow, Orange) and (Yellow, Green) challenge pairs, as the relative angle between these vectors is persistently 22.5° .

What about the antagonistic (Red, Green) challenge? Here, the target outcome is for Alice and Bob to output the **Opposite** bit. The angular separation between the Red basis and the Green basis is exactly $3 \times 22.5^\circ = 67.5^\circ$. The probability of Bob measuring the *same* bit is $\cos^2(67.5^\circ)$. Consequently, the probability of Bob measuring the *opposite* bit is the orthogonal complement:

$$\begin{aligned}P(\text{Opposite}) &= 1 - \cos^2(67.5^\circ) \\ &= \sin^2(67.5^\circ)\end{aligned}$$

Due to the complementary angles ($90^\circ - 67.5^\circ = 22.5^\circ$), we know that $\sin(67.5^\circ) = \cos(22.5^\circ)$. Therefore:

$$P(\text{Opposite}) = \cos^2(22.5^\circ) \approx 85.3\%$$

In all four equally probable scenarios, the quantum strategy guarantees success with a probability of roughly 85.3%.

Because $85.3\% > 75\%$, this experiment provides a mathematically irrefutable **Bell Inequality Violation**. It rigorously proves that the underlying mechanisms of the universe cannot be modeled by any theory of local hidden variables. The universe is fundamentally, structurally non-local.

9.5 Historical and Experimental Verification

The theoretical purity of this distinction sparked a race to execute the CHSH experiment in reality, transitioning it from mathematical abstraction to empirical law.

- **1982 - Alain Aspect:** Conducted the first rigorous Bell tests. He separated entangled photons by 13 meters (40 light-nanoseconds) and executed measurement basis switches dynamically within a 5 nanosecond window. This successfully closed the locality loophole, achieving an empirical success rate of approximately 84%, nearly matching the quantum theoretical prediction.
- **2015 - Ronald Hanson's Lab (Delft University):** Executed an exhaustive “loophole-free” Bell test. By utilizing entangled electron spins localized in diamond defects, and placing the systems 1300 meters apart across the university campus, they performed 294 rigorous experimental trials. The empirical success rate was unequivocally higher than the 75% classical limit. With a statistical p-value of 0.039, they comprehensively rejected the null hypothesis of local hidden variables.

Quantum mechanics stands entirely vindicated. These spooky correlations are a demonstrable reality of our universe. Equipped with this foundational proof, the subsequent architecture of quantum computation—including quantum teleportation, quantum cryptography, and algorithmic acceleration—rests upon empirically unshakeable ground.

10 Lecture 8: The No-Cloning Theorem and Quantum Teleportation

10.1 Introduction: Limits of Quantum Information

Our previous analysis of the CHSH game demonstrated a definitive quantum advantage, proving that entangled systems possess non-local correlations fundamentally impossible to replicate with classical hidden variables. To further our understanding of quantum computation, we must equally investigate the strict theoretical limitations imposed by quantum mechanics. Clarifying what is mathematically prohibited not only defines the boundaries of quantum information processing but directly motivates the development of ingenious protocols such as quantum teleportation. We begin with a foundational negative result known as the No-Cloning Theorem.

10.2 The No-Cloning Theorem

In classical computer science, copying information is a trivial operation. One can easily construct a logical circuit (such as a simple fan-out gate) that accepts a single input bit and deterministically outputs arbitrarily many identical copies. However, in 1982, Wootters and Zurek published a paper in *Nature* demonstrating that a universal copying machine is fundamentally impossible in the quantum realm.

The **No-Cloning Theorem** dictates that there exists no physical device, described by any valid quantum evolution, capable of taking an arbitrary, unknown qubit state $|\psi\rangle$ and producing two independent, unentangled copies $|\psi\rangle \otimes |\psi\rangle$.

It is critical to be precise regarding the scope of this theorem. It strictly forbids a *universal* input maker—a single physical machine that successfully clones *all* possible qubit states $|\psi\rangle$ simultaneously.

If an input state is strictly promised to belong to a set of perfectly distinguishable, orthogonal states (e.g., exclusively $|0\rangle$ or $|1\rangle$), a specialized cloner can be constructed simply by measuring the state in the computational basis and subsequently operating a macroscopic “zero-maker” or

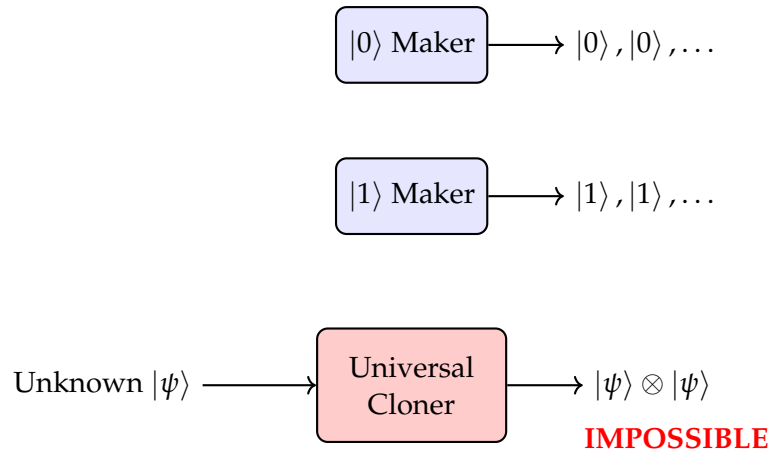


Figure 21: The distinction between physically permissible state generators and the prohibited universal quantum cloner. Constructing a machine that persistently generates a specific, known state (e.g., a laser polarized to output $|0\rangle$) is trivial. However, a generic “input maker” mapping an unknown $|\psi\rangle \mapsto |\psi\rangle^{\otimes 2}$ violates quantum mechanics.

“one-maker”. But if the state is an arbitrary superposition $\alpha |0\rangle + \beta |1\rangle$, any such measurement irrevocably collapses the state, permanently destroying the continuous amplitudes α and β .

A profound corollary of this theorem is the **unlearnability of a quantum state from a single copy**. If one possesses a single particle in an unknown state $|\psi\rangle$, it is physically impossible to determine the complex numbers α and β with perfect accuracy. If one could learn them, one could subsequently initialize a machine to generate infinite copies, thereby circumventing the No-Cloning restriction.

10.3 The Failure of the CNOT Gate as a Cloner

To build intuition regarding this impossibility, consider the Controlled-NOT (CNOT) gate, which represents the closest quantum analogue to classical copying. If we possess an unknown state $|\psi\rangle$ and initialize a secondary target ancilla in the state $|0\rangle$, does the CNOT gate successfully copy the amplitudes?

$$\begin{array}{l} \text{Input: } |\psi\rangle = \alpha |0\rangle + \beta |1\rangle \\ \text{Ancilla: } |0\rangle \end{array} \quad \begin{array}{c} \bullet \\ \hline \oplus \end{array} \quad \left. \vphantom{\begin{array}{c} \bullet \\ \hline \oplus \end{array}} \right\} \alpha |00\rangle + \beta |11\rangle$$

Figure 22: Attempting to clone an unknown superposition using a CNOT gate yields a highly entangled EPR-like state, rather than the separable cloned state $|\psi\rangle \otimes |\psi\rangle$.

The initial joint state of the system is:

$$\begin{aligned} |\Psi_{\text{in}}\rangle &= (\alpha |0\rangle + \beta |1\rangle) \otimes |0\rangle \\ &= \alpha |00\rangle + \beta |10\rangle \end{aligned}$$

Applying the linear CNOT transformation strictly flips the target qubit if the control is $|1\rangle$. Thus, the post-gate state evaluates to:

$$\text{CNOT} |\Psi_{\text{in}}\rangle = \alpha |00\rangle + \beta |11\rangle$$

We must compare this generated state against the theoretical ideal of two independent copies of

$|\psi\rangle$:

$$\begin{aligned} |\psi\rangle \otimes |\psi\rangle &= (\alpha |0\rangle + \beta |1\rangle) \otimes (\alpha |0\rangle + \beta |1\rangle) \\ &= \alpha^2 |00\rangle + \alpha\beta |01\rangle + \alpha\beta |10\rangle + \beta^2 |11\rangle \end{aligned}$$

The actual output $(\alpha |00\rangle + \beta |11\rangle)$ is a highly entangled state. It entirely lacks the cross-terms $(\alpha\beta |01\rangle$ and $\alpha\beta |10\rangle)$ required for the target tensor product. Notice that the ideal clone demands quadratic terms (α^2, β^2) . Because all isolated quantum evolutions are governed by unitary matrices—which are strictly linear transformations—it is mathematically impossible to dynamically square the input amplitudes.

10.4 Formal Proof of the No-Cloning Theorem

We can formalize this impossibility through a rigorous proof by contradiction. The proof rules out any arbitrary linear operation, even those employing arbitrary numbers of ancilla qubits and discarding unentangled residual “garbage”.

Assume, for the sake of contradiction, that there exists a universal unitary cloning machine U . It accepts an input state $|\psi\rangle$ alongside N ancilla qubits initialized to $|\bar{0}\rangle \equiv |0 \dots 0\rangle$. It flawlessly outputs two copies of $|\psi\rangle$ and some residual, unentangled garbage state $|g_\psi\rangle$:

$$U(|\psi\rangle \otimes |\bar{0}\rangle) = |\psi\rangle \otimes |\psi\rangle \otimes |g_\psi\rangle$$

Since this operator must work for all conceivable states simultaneously, we assume it correctly clones the computational basis states $|0\rangle$ and $|1\rangle$:

$$U(|0\rangle \otimes |\bar{0}\rangle) = |00\rangle \otimes |g_0\rangle \quad (16)$$

$$U(|1\rangle \otimes |\bar{0}\rangle) = |11\rangle \otimes |g_1\rangle \quad (17)$$

Additionally, it must correctly clone the uniform superposition state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$:

$$U(|+\rangle \otimes |\bar{0}\rangle) = |++\rangle \otimes |g_+\rangle \quad (18)$$

Because U is fundamentally a linear operator, we can substitute the definition of $|+\rangle$ into the left side of equation 18 and distribute the unitary transformation over the superposition:

$$U\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\bar{0}\rangle\right) = \frac{1}{\sqrt{2}}U(|0\rangle \otimes |\bar{0}\rangle) + \frac{1}{\sqrt{2}}U(|1\rangle \otimes |\bar{0}\rangle)$$

Substituting our established identities from equations 16 and 17 yields the *actual output* dictated by the laws of linear algebra:

$$\text{Actual Output} = \frac{1}{\sqrt{2}}(|00\rangle \otimes |g_0\rangle) + \frac{1}{\sqrt{2}}(|11\rangle \otimes |g_1\rangle)$$

We must now contrast this resulting vector against the hypothesized *target output* specified by equation 18:

$$\begin{aligned} \text{Target Output} &= |+\rangle \otimes |+\rangle \otimes |g_+\rangle \\ &= \left(\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle\right) \otimes |g_+\rangle \end{aligned}$$

These two state vectors are mathematically irreconcilable. The actual output produced by the linear transformation is an entangled state strictly lacking any amplitude in the $|01\rangle$ and $|10\rangle$ subspaces. A partial measurement of the first two qubits of the actual output yields 00 or 11 with a probability of 50% each. A partial measurement of the target output yields all four computational outcomes with a probability of 25% each.

Because a single physical operation cannot output two macroscopically distinct probability distributions simultaneously, the assumption that U exists results in an absolute contradiction. Therefore, universal quantum cloning is impossible. ■

10.5 Classical Analogy: The One-Time Flippable Coin

To intuitively contextualize this limitation, consider a classical scenario involving a biased coin. Suppose you are handed a single coin with a completely unknown bias—it might land heads 80% of the time, or perhaps 15%. Your task is to build a physical machine that takes this coin and manufactures two new coins possessing the exact same unknown bias.

If you are strictly restricted to flipping the original coin *only one time*, the cloning task is structurally impossible. A single flip yields exactly one classical bit of information (Heads or Tails). This provides virtually zero statistical data to reverse-engineer the underlying continuous real numbers governing the probability distribution.

A quantum state behaves as the ultimate physical manifestation of this “one-time flippable coin”. The process of quantum measurement destructively collapses the delicate superposition, yielding a single discrete classical bit while permanently deleting the original continuous amplitudes.

However, if one had access to an ensemble of n identical copies of the unknown quantum state $(|\psi\rangle^{\otimes n})$, one could perform repeated measurements to estimate the underlying amplitudes with precision proportional to $1/\sqrt{n}$. In quantum computer science, this systematic learning procedure is formally known as **Quantum State Tomography**. But from a single isolated copy, the continuous information is irrevocably locked behind the measurement collapse.

10.6 Quantum Teleportation

The harsh restrictions imposed by the No-Cloning Theorem initially suggest that transferring unknown quantum information between physically separated systems is insurmountably fragile. However, in 1993, Bennett, Brassard, and colleagues demonstrated a profound protocol capable of circumventing these apparent limitations: **Quantum Teleportation**.

Imagine Alice sits in her laboratory in Gates Hall, possessing an unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. She wishes to transfer this precise quantum state to her friend Bob, located across campus in Wean Hall, without physically transporting the delicate particle. To achieve this, Alice and Bob must exploit a pre-shared entangled resource: an EPR pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, commonly referred to as an *e-bit*, distributed between their locations ahead of time.

The foundational paradigm of quantum teleportation is summarized by the resource equation:

$$1 \text{ } e\text{-bit (EPR Pair)} + 2 \text{ Classical Bits} \equiv 1 \text{ Transmitted Qubit}$$

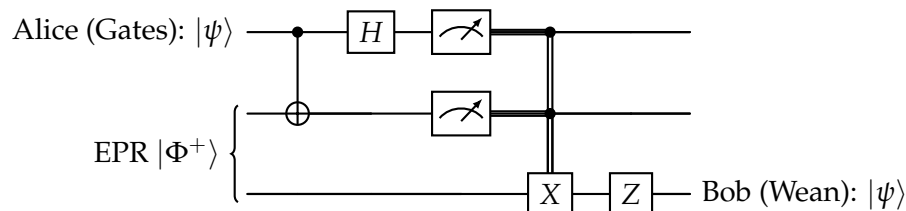


Figure 23: The standard Quantum Teleportation circuit. Alice performs a joint Bell-basis measurement between her unknown state and her half of the EPR pair. She transmits two classical bits to Bob, instructing him to apply conditional Pauli corrections (X and Z) to reconstruct $|\psi\rangle$.

10.6.1 Step 1: The Magical Distributed CNOT

To pedagogically decode the mechanics, let us conceptually divide the protocol. Imagine Alice and Bob possessed a magical ability to execute a *distributed* CNOT gate between their spatially

separated laboratories. If Alice controls the CNOT using her unknown state $|\psi\rangle$, and Bob provides a fresh $|0\rangle$ target, the joint state evolves into an entangled distribution identical to our failed cloning attempt:

$$\text{Magic CNOT}((\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle) = \alpha|00\rangle + \beta|11\rangle$$

This state effectively maps the continuous amplitudes α and β onto a joint entangled space. To compress the quantum information strictly into Bob's laboratory, Alice applies a local Hadamard gate to her qubit and measures it.

$$\begin{aligned} (H \otimes I)(\alpha|00\rangle + \beta|11\rangle) &= \alpha \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle + \beta \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle \\ &= \frac{1}{\sqrt{2}} [|0\rangle (\alpha|0\rangle + \beta|1\rangle) + |1\rangle (\alpha|0\rangle - \beta|1\rangle)] \end{aligned}$$

Alice measures her qubit in the standard basis.

- With 50% probability, she measures 0. Bob's state instantaneously collapses to $\alpha|0\rangle + \beta|1\rangle$. Bob natively holds the perfect state $|\psi\rangle$.
- With 50% probability, she measures 1. Bob's state collapses to $\alpha|0\rangle - \beta|1\rangle$. To resolve this phase error, Alice texts Bob the classical bit '1'. Bob subsequently applies a Phase gate (Z-gate), flipping the negative sign and precisely recovering $|\psi\rangle$.

10.6.2 Step 2: Synthesizing the Distributed CNOT via Entanglement

The magical distributed CNOT violates physical constraints on non-local operations. However, Alice and Bob can synthesize its exact effects entirely locally by sacrificing their pre-shared EPR pair.

The initial joint state of the three-particle system (Alice's unknown $|\psi\rangle$, Alice's half of the EPR pair, and Bob's half) is:

$$\begin{aligned} |\Psi_1\rangle &= |\psi\rangle \otimes |\Phi^+\rangle \\ &= (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle) \end{aligned}$$

Alice performs a strictly local CNOT inside her own office, using $|\psi\rangle$ as the control and her half of the EPR pair as the target. Since the control is strictly the first qubit, the third (Bob's) qubit is untouched:

$$|\Psi_2\rangle = \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle)$$

Next, Alice locally measures her second particle (her half of the EPR pair). Let us calculate the outcome probabilities by grouping the state terms according to the value of the middle qubit:

$$|\Psi_2\rangle = |\cdot 0 \cdot\rangle \left(\frac{\alpha}{\sqrt{2}} |00\rangle + \frac{\beta}{\sqrt{2}} |11\rangle \right) + |\cdot 1 \cdot\rangle \left(\frac{\alpha}{\sqrt{2}} |01\rangle + \frac{\beta}{\sqrt{2}} |10\rangle \right)$$

- If Alice measures 0, the probability is $P(0) = \left| \frac{\alpha}{\sqrt{2}} \right|^2 + \left| \frac{\beta}{\sqrt{2}} \right|^2 = \frac{1}{2}(|\alpha|^2 + |\beta|^2) = \frac{1}{2}$. The joint state of Alice's remaining qubit (qubit 1) and Bob's qubit (qubit 3) perfectly collapses to $\alpha|00\rangle + \beta|11\rangle$. This is the exact output of the magical distributed CNOT!

- If Alice measures 1, the probability is identically $1/2$. The remaining system collapses to $\alpha|01\rangle + \beta|10\rangle$. This state is inverted. Alice simply texts Bob her measurement outcome. Upon receiving the classical text '1', Bob locally applies an X gate (quantum NOT) to his particle, physically mapping the state to $\alpha|00\rangle + \beta|11\rangle$.

By locally performing a CNOT, measuring, and communicating one classical bit, the distributed CNOT has been synthesized. Alice then immediately executes the Hadamard operation and second measurement on her remaining qubit, sending the final classical bit to Bob to trigger the conditional Z correction.

10.7 Concluding Observations and Entanglement Swapping

At the conclusion of the protocol, the exact quantum information constituting $|\psi\rangle$ resides exclusively in Bob's laboratory. Alice's original state has been destructively measured into classical basis states, thereby guaranteeing strict adherence to the No-Cloning Theorem. Furthermore, because the entire transfer dynamically stalls until Bob receives and processes the light-speed-limited classical texts, relativity and physical causality are perfectly preserved.

Remarkably, teleportation operates completely agnostically to the source of the input state. Suppose a third party, Charlie, is located in Doherty Hall. If Alice prepares an EPR pair and teleports one half of it to Bob, and the other half to Charlie, Bob and Charlie will find themselves sharing a maximally entangled EPR pair despite having never physically interacted. This profound extension is known as **Entanglement Swapping**, establishing that entanglement itself can be manipulated and distributed strictly through quantum teleportation networks.

11 Lecture 9: Quantum Money and Cryptography

11.1 Historical Context and the No-Cloning Advantage

In classical computer science, the fundamental property of digital information is its strict ability to be copied indefinitely. A binary string etched onto a magnetic hard drive or printed on a physical substrate can be duplicated with perfect fidelity, bounding the security of classical banknotes or subway tokens to the macroscopic difficulty of reproducing complex physical watermarks.

However, in 1969—more than a decade before physicists like Richard Feynman formally initiated the field of quantum computation—Stephen Wiesner, an undergraduate at Brandeis University, proposed a cryptographic scheme that leveraged the fundamental unlearnability of quantum states to construct unforgeable currency. Although his paper was initially rejected as “nonsense” by the IEEE Transactions on Information Theory, the concept survived through his collaborator Charles Bennett. Wiesner's intuition predated and perfectly aligned with the **No-Cloning Theorem** (Wootters and Zurek, 1982), which rigorously proved that no physical unitary operation can duplicate an arbitrary, unknown quantum state $|\psi\rangle$ into $|\psi\rangle \otimes |\psi\rangle$.

Because unknown quantum states cannot be cloned, they constitute the ultimate theoretical substrate for money. While holding coherent states in a physical “quantum wallet” remains beyond current decoherence limits, the underlying mechanics of this protocol directly birthed the field of *Quantum Key Distribution* (QKD), an unconditionally secure cryptographic protocol actively deployed in commercial optical networks today.

11.2 Wiesner's Quantum Money Protocol

To instantiate the quantum money scheme, we define a central trusted authority—the Bank (or Transit Authority, in the subway token model). The protocol relies on a security parameter n (e.g., $n = 256$), which dictates the exponential difficulty of counterfeiting.

1. Generation Phase: When the bank mints a new quantum coin, it executes the following steps:

1. Generates a random classical serial number $s \in \{0, 1\}^n$.
2. Generates a secret quantum description string $q \in \{0, 1, +, -\}^n$. To avoid maintaining a massive, centralized ledger of every minted coin, the bank may utilize a cryptographic pseudo-random function (PRF) keyed with a master secret key k . Thus, $q = \text{PRF}_k(s)$.
3. Prepares an n -qubit quantum state $|\psi\rangle$ corresponding to the string q . For example, if $q = (0, +, 1, -, \dots)$, the physical state prepared is:

$$|\psi\rangle = |0\rangle \otimes |+\rangle \otimes |1\rangle \otimes |-\rangle \dots$$

The final minted coin given to the user is the physical pair $(s, |\psi\rangle)$.

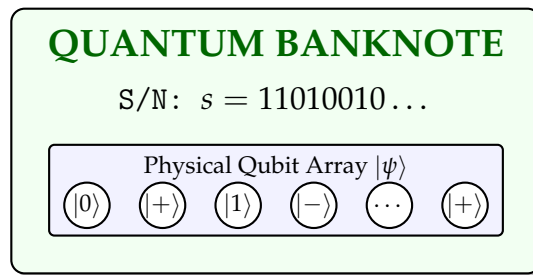


Figure 24: Conceptual diagram of a Wiesner Quantum Banknote. The classical serial number s is publicly legible, while the multi-qubit state $|\psi\rangle$ remains an unknown, unclonable quantum state from the user's perspective.

2. Verification Phase: When a user attempts to spend the coin, they return the physical pair $(s, |\psi\rangle)$ to the bank.

1. The bank reads the classical serial number s and recomputes the secret string $q = \text{PRF}_k(s)$.
2. For each qubit $i \in \{1, \dots, n\}$, the bank measures the particle in the correct orthogonal basis defined by q_i . If $q_i \in \{0, 1\}$, it measures in the standard basis $\{|0\rangle, |1\rangle\}$. If $q_i \in \{+, -\}$, it measures in the diagonal (Hadamard) basis $\{|+\rangle, |-\rangle\}$.
3. If all n measurement outcomes perfectly match the elements of q , the coin is verified as genuine. If even a single measurement yields an orthogonal outcome, the coin is immediately confiscated, and an alarm is triggered.

11.3 Naive Counterfeiting: The $\frac{5}{8}$ Attack

Let us analyze the success probability of a malicious user attempting to turn one valid coin into two valid coins. By the No-Cloning Theorem, the counterfeiter cannot perfectly duplicate $|\psi\rangle$. A naive classical strategy is to measure the entire state in the standard basis $\{|0\rangle, |1\rangle\}$, obtain a classical string $r \in \{0, 1\}^n$, and synthesize two fresh sets of qubits in the state $|r\rangle$.

We must evaluate the probability that *both* forged coins pass the bank's verification procedure. We analyze a single qubit i :

- **Case 1: The true state is computational (Probability $1/2$).** The true state is $|0\rangle$ or $|1\rangle$. Measuring in the standard basis yields the correct classical bit with 100% accuracy. The counterfeiter produces two identical, correct copies. The bank will measure both in the standard basis. Both pass with probability $1 \times 1 = 1$.

- **Case 2: The true state is diagonal (Probability 1/2).** The true state is $|+\rangle$ or $|-\rangle$. The counterfeiter measures in the standard basis, inherently collapsing the state to a random bit $r_i \in \{0,1\}$ with a 50/50 distribution. The counterfeiter then places $|r_i\rangle$ into both forged coins. The bank expects $|+\rangle$ or $|-\rangle$, and thus measures both forged coins in the diagonal basis. Measuring a standard basis state in the diagonal basis yields the expected outcome with exactly 50% probability. Crucially, the bank tests both coins independently. The probability that the first coin passes is 1/2. The probability that the second coin passes is 1/2. The joint probability of both passing is 1/4.

Averaging over the bank's secret basis choices, the success probability for a single qubit $P_i(\text{success})$ is:

$$\begin{aligned} P_i(\text{success}) &= \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{4}\right) \\ &= \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 62.5\% \end{aligned}$$

Because the states are tensor-product independent, the total probability of simultaneously passing the verification for all n qubits is exponentially small:

$$P_{\text{total}} = \left(\frac{5}{8}\right)^n$$

For $n = 256$, this probability evaluates to approximately 2^{-173} , a thermodynamically negligible number. (Note: The theoretically optimal independent-measurement attack achieves a bound of exactly $(\frac{3}{4})^n$, which remains physically intractable).

11.4 Interactive Counterfeiting: The Bomb-Tester Attack

The security of Wiesner's money heavily relies on an implicit assumption regarding the bank's behavior post-verification. Suppose the bank implements a "user-friendly" policy: if a coin successfully passes verification, the bank simply returns the collapsed post-measurement state to the user. Since a correct measurement does not disturb the state (e.g., measuring $|+\rangle$ in the diagonal basis yields $|+\rangle$), returning a valid coin seemingly preserves the system.

However, Nagaj et al. (2011) proved that returning a verified coin opens a fatal vulnerability. A counterfeiter can execute an **Interactive Attack**—conceptually isomorphic to the Elitzur-Vaidman Bomb Tester—to extract the full classical description q of the state using $\mathcal{O}(n^2)$ verification queries, circumventing the No-Cloning Theorem entirely.

The attack targets one qubit $|u\rangle$ at a time (holding the other $n - 1$ qubits in their correct, verified state to avoid extraneous alarms). The counterfeiter wishes to determine if $|u\rangle$ is precisely $|+\rangle$.

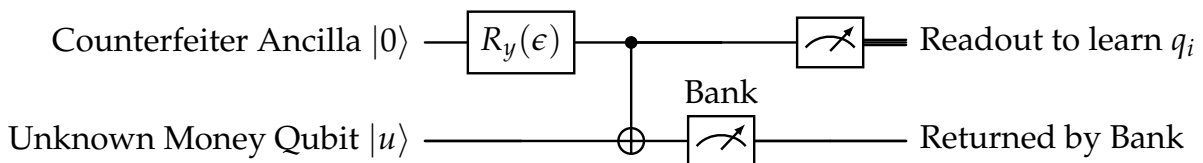


Figure 25: A single iteration of the Interactive Counterfeiting attack. The counterfeiter gently entangles an ancilla with the money qubit before passing it to the naive bank. The bank's measurement acts as the quantum Zeno collapse.

The counterfeiter defines a large integer C and an infinitesimal angle $\epsilon = \frac{\pi}{2C}$. The protocol proceeds iteratively:

1. The counterfeiter initializes a local ancilla qubit in state $|0\rangle$.
2. They rotate the ancilla by ϵ :

$$|\psi_{\text{ancilla}}\rangle = \cos(\epsilon) |0\rangle + \sin(\epsilon) |1\rangle$$

3. They apply a CNOT gate, using their ancilla as the control and the unknown money qubit $|u\rangle$ as the target.
4. They submit the modified money qubit to the bank for verification.

Let us rigorously analyze the joint wave-function under the two possible cases for the true state of $|u\rangle$:

Case 1: The true state is $|+\rangle$. The state prior to the CNOT is $(\cos \epsilon |0\rangle + \sin \epsilon |1\rangle) \otimes |+\rangle$. Applying the CNOT flips the target strictly when the control is 1. However, the Pauli-X operator acting on the $|+\rangle$ state is an invariant identity operation: $X|+\rangle = |+\rangle$.

$$\begin{aligned} \text{CNOT}(\cos \epsilon |0\rangle |+\rangle + \sin \epsilon |1\rangle |+\rangle) &= \cos \epsilon |0\rangle |+\rangle + \sin \epsilon |1\rangle (X|+\rangle) \\ &= \cos \epsilon |0\rangle |+\rangle + \sin \epsilon |1\rangle |+\rangle \\ &= (\cos \epsilon |0\rangle + \sin \epsilon |1\rangle) \otimes |+\rangle \end{aligned}$$

The CNOT gate utterly fails to entangle the qubits. When the bank receives the target, it measures in the diagonal basis (expecting $|+\rangle$) and observes $+$ with 100% probability. The bank does not call the cops and returns the $|+\rangle$ state unharmed. Meanwhile, the counterfeiter's ancilla remains uncollapsed, successfully retaining its ϵ rotation. If the counterfeiter repeats this loop C times, the ancilla continuously rotates by ϵ until it reaches exactly $C \times (\frac{\pi}{2C}) = \frac{\pi}{2}$ radians, deterministically arriving at the state $|1\rangle$.

Case 2: The true state is $|0\rangle$. The state prior to the CNOT is $(\cos \epsilon |0\rangle + \sin \epsilon |1\rangle) \otimes |0\rangle$. The CNOT successfully entangles the system:

$$\text{CNOT}(\cos \epsilon |0\rangle |0\rangle + \sin \epsilon |1\rangle |0\rangle) = \cos \epsilon |00\rangle + \sin \epsilon |11\rangle$$

The bank receives the target and measures in the standard basis (expecting $|0\rangle$).

- With probability $\sin^2(\epsilon)$, the bank observes 1. Verification fails, and the cops are called.
- With probability $\cos^2(\epsilon)$, the bank observes 0. The wave-function instantaneously collapses the counterfeiter's ancilla back to $|0\rangle$.

If the cops are not called, the system resets. The rotation does *not* accumulate. Over C iterations, the total probability of the cops being called is union-bounded by:

$$P(\text{Cops}) \leq C \cdot \sin^2(\epsilon) \approx C \cdot \epsilon^2 = C \left(\frac{\pi}{2C} \right)^2 = \frac{\pi^2}{4C}$$

For a sufficiently large C , the probability of getting caught approaches 0. The ancilla is persistently collapsed back to $|0\rangle$.

11.5 Conclusion: The Necessity of Reissuance

At the end of the C iterations, the counterfeiter simply measures their own local ancilla in the standard basis.

- If the readout is 1, they know with absolute certainty that the target qubit was $|+\rangle$.
- If the readout is 0, they know the target qubit was *not* $|+\rangle$.

By systematically testing the four possible states ($\{0, 1, +, -\}$) for each of the n qubits, the counterfeiter perfectly extracts the secret classical string q using $\mathcal{O}(n)$ tests per qubit ($\mathcal{O}(n^2)$ total queries), while maintaining an arbitrarily low probability of detection.

This profound vulnerability demonstrates that the bank **must never** return a verified coin to the user. To maintain cryptographic security against interactive quantum Zeno attacks, a verifying authority must operate strictly destructively: upon successful verification, the bank must cross the serial number s off its ledger, throw the physical state away, and mint an entirely new quantum coin to hand back to the user.

12 Lecture 10: Classical Simulation and Reversible Computation

12.1 The Classical Circuit Model and Computational Efficiency

To rigorously ground the computational power of quantum mechanics, we must first establish a formal baseline using the classical circuit model of computation. In this framework, we evaluate the complexity of computing a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ via acyclic networks of fundamental logical gates, typically the universal set $\{\text{AND}, \text{OR}, \text{NOT}\}$.

When analyzing algorithmic efficiency, our primary metric is the asymptotic scaling of the number of gates with respect to the input size n . While theoretical computer science often relies on the Turing machine model or high-level programming paradigms (e.g., Python or C), the circuit model maps elegantly to these standard abstractions. Specifically, if a deterministic algorithm executes in T time steps on an n -bit input, one can construct an equivalent Boolean circuit containing $\mathcal{O}(T \log T)$ gates. This tight correspondence justifies using circuit size as our fundamental measure of computational time.

In 1937, Claude Shannon established strict theoretical limits on the efficiency of Boolean circuits. Shannon demonstrated that any arbitrary Boolean function can be computed using at most $\mathcal{O}(2^n/n)$ gates. However, he concurrently proved a pessimistic lower bound: *almost all* possible Boolean functions require at least $\mathcal{O}(2^n/n)$ gates. While evaluating a generic random function is therefore physically intractable for large n , computer science focuses almost exclusively on the vanishingly small subset of structured functions (e.g., multiplication, primality testing, shortest paths) that can be computed with a physically plausible *polynomial* number of gates, $\mathcal{O}(n^k)$.

12.2 Probabilistic Computation

Because quantum mechanics is inherently probabilistic upon measurement, a strictly deterministic classical baseline is an insufficient point of comparison. We must augment our classical circuits with access to randomness, defining the probabilistic computing paradigm (analogous to the BPP complexity class).

We augment the standard model with a “Coin Flip” gate—a primitive with no inputs and one output wire that evaluates to 0 with probability $1/2$ and 1 with probability $1/2$. A probabilistic circuit C successfully computes a Boolean function $f(x)$ if, for all inputs $x \in \{0, 1\}^n$, the probability of the circuit outputting an incorrect answer over the internal coin flips is strictly bounded:

$$\Pr[C(x) \neq f(x)] \leq \epsilon$$

where $\epsilon < 1/2$. While probabilistic algorithms can yield polynomial reductions in circuit size, complexity theorists strongly conjecture that probabilistic computation does not provide exponential speedups over deterministic classical computers (i.e., $P = BPP$).

12.3 The Quantum Circuit Model

The architecture of a quantum computer directly mirrors the classical circuit paradigm but operates on the continuous state space of complex amplitudes. A quantum circuit evaluating an n -bit input x fundamentally proceeds in three phases:

1. **Initialization:** The n input qubits are initialized to the unentangled classical basis state corresponding to the string x : $|x\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$.
2. **Evolution:** The system is subjected to a sequence of unitary transformations (e.g., CNOT, Hadamard, and phase rotations), forming one monolithic global unitary operator U acting on $\mathbb{C}^{2^n}$.
3. **Measurement:** The final quantum state is measured in the standard computational basis to extract classical information.

The Principle of Deferred Measurement: In physical implementations, measurement operations might occur intermediately, dynamically dictating future gate applications based on classical outcomes. However, the formal Principle of Deferred Measurement guarantees that any quantum circuit containing intermediate measurements can be mathematically and operationally transformed into an equivalent circuit where all measurements are strictly postponed to the absolute end, absorbing classical conditionality into coherent quantum controlled operations.

12.4 Reversibility, Landauer's Principle, and Thermodynamics

Before attempting to simulate a classical circuit with a quantum one, we immediately encounter a foundational physical asymmetry. The allowable gates in quantum mechanics are strictly unitary operators U , which inherently obey the identity:

$$U^\dagger U = I$$

This dictates that every quantum transformation is perfectly invertible ($U^{-1} = U^\dagger$). If one knows the complete output state, one can deterministically reverse the temporal evolution to recover the exact input state.

Conversely, fundamental classical gates such as the AND gate are irreversibly destructive. The inputs $(0, 0)$, $(0, 1)$, and $(1, 0)$ all map to the identical output 0. The mapping is non-injective; given an output of 0, it is mathematically impossible to logically invert the gate and uniquely determine the inputs. The gate fundamentally erases information.

In the 1960s, physicist Rolf Landauer formalized the thermodynamic consequences of logical irreversibility. Landauer's Principle posits that any logically irreversible operation (like an AND gate) inevitably results in an entropic increase in the environment, mandating the dissipation of heat. Specifically, erasing one bit of information radiates at least $k_B T \ln 2$ Joules of thermal energy. In contrast, fully reversible computations can theoretically be executed in a closed thermodynamic system with zero energy loss. Because quantum evolution is dictated by unitary (reversible) matrices, quantum computation is inherently a zero-dissipation process—provided no measurements (which are irreversible collapses) occur.

12.5 The Controlled-SWAP (Fredkin) Gate

To prove that a quantum computer is at least as powerful as a classical one, we must demonstrate that unitary quantum operations can flawlessly simulate standard, irreversible classical gates (AND, OR). To bridge this gap, we introduce a universal reversible gate: the Controlled-SWAP (CSWAP), or Fredkin gate.

The CSWAP is a three-qubit gate. It designates the first qubit as the control, and conditionally swaps the states of the second and third target qubits strictly if the control is $|1\rangle$.

$$\begin{aligned}\text{CSWAP } |0, x_2, x_3\rangle &= |0, x_2, x_3\rangle \\ \text{CSWAP } |1, x_2, x_3\rangle &= |1, x_3, x_2\rangle\end{aligned}$$

Because it acts purely as a deterministic permutation of the eight canonical computational basis states, the CSWAP matrix is an orthogonal permutation matrix. Consequently, it is unitary and represents a perfectly valid, physically realizable quantum operation. In hardware, a CSWAP can be decomposed into a network of CNOT and Toffoli gates. Furthermore, the CSWAP is its own inverse; applying it twice sequentially yields the identity I .

12.5.1 Simulating Classical Logic via CSWAP

By injecting supplementary constant inputs—known as **ancilla bits**—we can coax the CSWAP gate into evaluating non-reversible Boolean logic while preserving global reversibility.

1. Simulating the AND Gate: We inject the bits x_1 and x_2 alongside a static ancilla bit initialized to $|0\rangle$.

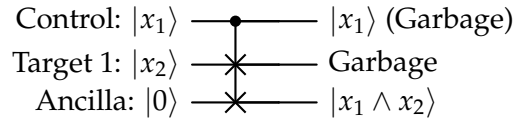


Figure 26: Simulating a classical AND gate using a reversible CSWAP gate and a $|0\rangle$ ancilla.

Let us trace the bottom output wire. If $x_1 = 0$, no swap occurs, and the $|0\rangle$ ancilla passes through unmodified. If $x_1 = 1$, the swap triggers, routing the value of x_2 down to the bottom wire. In Boolean logic, this perfectly mirrors $x_1 \wedge x_2$. The other wires output residual states that we do not care about; these are termed **garbage bits**.

2. Simulating Fan-Out (Duplication): Classical circuits arbitrarily branch wires to duplicate signals. The No-Cloning theorem forbids copying arbitrary superpositions, but we only require copying classical basis states $\{|0\rangle, |1\rangle\}$ for standard classical simulation. We construct a fan-out gate by placing the value to be copied on the control wire, an ancilla $|1\rangle$ on Target 1, and an ancilla $|0\rangle$ on Target 2.

- If Control = 0, no swap occurs, outputting (0, 1, 0).
- If Control = 1, the 1 and 0 swap, outputting (1, 0, 1).

The top wire and the bottom wire both perfectly echo the input value x_1 .

12.6 The Problem of Quantum Garbage

By cascading CSWAP operations, any classical circuit evaluating a generic Boolean function $f(x)$ can be converted into an equivalent reversible circuit requiring $\mathcal{O}(\text{poly}(n))$ gates and $\mathcal{O}(\text{poly}(n))$ ancilla bits. The resulting unitary transformation P operates as:

$$P(|x\rangle \otimes |\bar{0}\rangle) = |x\rangle \otimes |f(x)\rangle \otimes |g(x)\rangle$$

where $|\bar{0}\rangle$ denotes the sequence of initialized ancilla bits, and $|g(x)\rangle$ encompasses all the intermediate garbage bits.

If the sole purpose of the algorithm is to evaluate a classical function and read the terminal answer, this formulation suffices. However, in advanced quantum computing, we intend to

execute P not merely on isolated classical bitstrings, but on massive, coherent superpositions over all possible inputs: $\sum_x \alpha_x |x\rangle$. By linearity, passing a superposition through P yields:

$$\sum_x \alpha_x |x\rangle |f(x)\rangle |g(x)\rangle$$

In this domain, the garbage bits $|g(x)\rangle$ become cataclysmic. Because the garbage uniquely depends on the evaluation path of x , it acts as an implicit measurement record of the computation. The answer register $|f(x)\rangle$ remains highly entangled with the massive, unstructured $|g(x)\rangle$ register. If we subsequently attempt to perform wave interference on the $|f(x)\rangle$ outcomes, the orthogonal garbage states prevent the probability amplitudes from correctly overlapping and cancelling. Conceptually, ignoring or “trashing” the garbage constitutes taking a partial trace over the system, which mathematically collapses the pure superposition into an incoherent, mixed density matrix. Thus, a quantum subroutine must strictly clean up after itself.

12.7 Bennett’s Uncomputing Protocol

In 1989, Charles Bennett introduced an elegant protocol to systematically annihilate computational garbage, rendering reversible computation genuinely suitable for quantum superpositions.

The protocol requires an additional output register initialized to $|y\rangle$ (typically $|0\rangle$) specifically to hold the target answer, decoupled from the functional mechanism of the circuit. The sequence is as follows:

1. **Compute:** Execute the primary reversible circuit P to calculate $f(x)$ alongside its inevitable garbage.
2. **Copy:** Apply a sequence of CNOT gates using the $f(x)$ register as the control and the isolated $|y\rangle$ register as the target. This flips the target exactly to $y \oplus f(x)$.
3. **Uncompute:** Because unitary gates are invertible, apply the inverse operation $P^\dagger$ to the input, functional, and garbage registers.

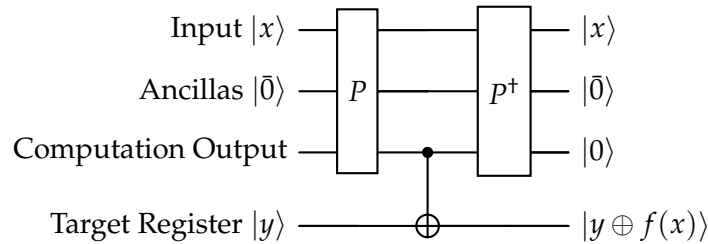


Figure 27: Bennett’s uncomputing trick. The classical answer is safely transferred to an isolated target register via CNOT. Since $P^\dagger P = I$, the reverse circuit perfectly untangles and collapses the garbage back down to pristine $|0\rangle$ ancillas.

Let us rigorously trace the state vector algebraically. Assume $y = 0$:

$$|\Psi_1\rangle = |x\rangle \otimes |\bar{0}\rangle \otimes |0\rangle \quad (\text{Initial State})$$

$$|\Psi_2\rangle = (P |x\rangle |\bar{0}\rangle) \otimes |0\rangle = |x\rangle |f(x), g(x)\rangle |0\rangle \quad (\text{Post-Compute})$$

$$|\Psi_3\rangle = \text{CNOT}(|x\rangle |f(x), g(x)\rangle |0\rangle) = |x\rangle |f(x), g(x)\rangle |f(x)\rangle \quad (\text{Post-Copy})$$

$$|\Psi_4\rangle = (P^\dagger |x\rangle |f(x), g(x)\rangle) \otimes |f(x)\rangle \quad (\text{Post-Uncompute})$$

Because $P^\dagger$ perfectly reverses P , the expression $P^\dagger |x\rangle |f(x), g(x)\rangle$ collapses mathematically strictly back to the initial state $|x\rangle |\bar{0}\rangle$. Consequently, the terminal state is:

$$|\Psi_{\text{final}}\rangle = |x\rangle \otimes |\bar{0}\rangle \otimes |f(x)\rangle$$

12.8 Formal Definition of Quantum Boolean Function Evaluation

By utilizing Bennett's uncomputing trick, we can package any classical computation into a perfectly clean, reversible unitary box. The internal ancilla bits start at $|0\rangle$ and return to $|0\rangle$, acting strictly as enclosed catalytic workspace memory. From an external algorithmic perspective, they can be factored out.

Thus, we adopt the following standard mathematical convention: A quantum circuit implements a classical Boolean function $f(x)$ if its corresponding unitary operator U_f maps computational basis states according to:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

where $\oplus$ signifies bitwise addition modulo 2 (XOR). When evaluating an arbitrary coherent superposition, the linearity of quantum mechanics guarantees that U_f evaluates the function across all basis states simultaneously—without generating entanglement-destroying garbage:

$$U_f \left(\sum_x \alpha_x |x\rangle \otimes |0\rangle \right) = \sum_x \alpha_x |x\rangle |f(x)\rangle$$

This garbage-free parallel mapping constitutes the supreme foundational engine of all advanced quantum algorithms.

13 Lecture 11: Revealing XOR-Patterns and the Bernstein-Vazirani Algorithm

13.1 Standard vs. Sign Implementation of Boolean Functions

In classical computer science, evaluating a boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ maps an n -bit input x to a single-bit output. When designing quantum algorithms, we must construct a reversible unitary operator that evaluates this function. The standard paradigm, as established previously, involves an input register, an output register, and a set of initialized ancilla qubits (which we typically assume are uncomputed and revert to $|0\rangle$).

The formal definition of implementing a boolean function f via a quantum circuit U_f maps the basis states as follows:

$$U_f |x\rangle \otimes |b\rangle = |x\rangle \otimes |b \oplus f(x)\rangle$$

where $\oplus$ denotes bitwise addition modulo 2 (the logical XOR operation).

If the target register b is initialized to $|0\rangle$, the output perfectly encodes the function evaluation: $|x\rangle |f(x)\rangle$. However, because quantum computation is linear, we can initialize the target register b in a superposition. Let us evaluate the behavior of the circuit when b is placed in the state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Plugging this into the unitary operator U_f , we apply linearity:

$$\begin{aligned} U_f \left(|x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= \frac{1}{\sqrt{2}} U_f |x, 0\rangle - \frac{1}{\sqrt{2}} U_f |x, 1\rangle \\ &= \frac{1}{\sqrt{2}} |x, 0 \oplus f(x)\rangle - \frac{1}{\sqrt{2}} |x, 1 \oplus f(x)\rangle \end{aligned}$$

We now analyze the two possible classical values for $f(x)$:

- **Case $f(x) = 0$:** The state evaluates to $\frac{1}{\sqrt{2}}(|x, 0\rangle - |x, 1\rangle) = |x\rangle \otimes |-\rangle$.
- **Case $f(x) = 1$:** The state evaluates to $\frac{1}{\sqrt{2}}(|x, 1\rangle - |x, 0\rangle) = -|x\rangle \otimes |-\rangle$.

Remarkably, the target register remains in the state $|-\rangle$ entirely unchanged. However, the overall wave-function acquires a conditional negative sign exactly when $f(x) = 1$. We can express this uniformly using the mathematical identity $(-1)^{f(x)}$:

$$U_f(|x\rangle \otimes |-\rangle) = (-1)^{f(x)} |x\rangle \otimes |-\rangle$$

This mechanism is known as **Phase Kickback** or the **Sign Implementation**. Because the target register $|-\rangle$ factors out and remains unentangled, we can logically omit it from our notation and define an effective quantum operation $Q_f^\pm$ acting strictly on the n -qubit input space:

$$Q_f^\pm |x\rangle = (-1)^{f(x)} |x\rangle$$

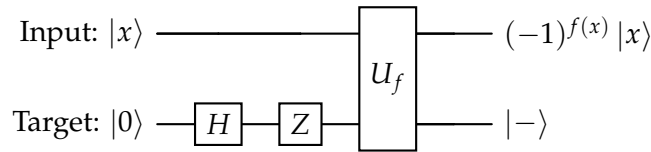


Figure 28: The Phase Kickback circuit. By applying a Hadamard and a Pauli-Z (or NOT followed by Hadamard) to the initialized $|0\rangle$ target, we prepare the $|-\rangle$ state. The standard evaluation of U_f subsequently projects the function's output into the phase of the input register.

13.2 Example: The 2-Bit XOR Function

Consider the simple 2-bit function $f(x_1, x_2) = x_1 \oplus x_2$, frequently referred to as the “not equals” function. The truth table is straightforward:

$$\begin{aligned} f(0,0) = 0 &\implies (-1)^0 = +1 \\ f(0,1) = 1 &\implies (-1)^1 = -1 \\ f(1,0) = 1 &\implies (-1)^1 = -1 \\ f(1,1) = 0 &\implies (-1)^0 = +1 \end{aligned}$$

Under the sign implementation paradigm, the quantum gate $Q_{x_1 \oplus x_2}^\pm$ corresponds strictly to a 4×4 diagonal unitary matrix operating on $\mathbb{C}^4$:

$$Q_{\text{XOR}}^\pm = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

13.3 Superposition and the Futility of Naive Measurement

The true power of the sign implementation manifests when we introduce quantum superpositions. Rather than passing a single classical string x , we prepare the **uniform superposition state** over all $N = 2^n$ classical strings. This state is reliably constructed by applying a layer of n Hadamard gates ($H^{\otimes n}$) to the initial ground state $|0\rangle^{\otimes n}$:

$$\begin{aligned} H^{\otimes n} |0 \dots 0\rangle &= |+\rangle \otimes |+\rangle \otimes \dots \otimes |+\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \end{aligned}$$

When we feed this uniform superposition into our sign-implementing circuit $Q_f^\pm$, linearity demands that the operator evaluates the function simultaneously across all parallel branches:

$$Q_f^\pm \left(\frac{1}{\sqrt{N}} \sum_x |x\rangle \right) = \frac{1}{\sqrt{N}} \sum_x (-1)^{f(x)} |x\rangle$$

This is conceptually profound: with a single application of the quantum circuit, the entire truth table of the boolean function $f(x)$ of size 2^n is encoded into the probability amplitudes of the resulting state.

However, one must immediately ask: *Is this state actually useful?* If we simply perform a standard measurement on this state, the probability of obtaining any specific string x^* is given by the Born rule:

$$P(x^*) = \left| \frac{(-1)^{f(x^*)}}{\sqrt{N}} \right|^2 = \frac{1}{N}$$

The global signs cancel upon squaring. The measurement apparatus will simply output a uniformly random string x , offering exactly zero insights regarding the encoded phase differences. Generating a random bitstring classically requires trivial effort, rendering this naive single-step quantum procedure effectively useless.

13.4 The Destructive Nature of Quantum Garbage

Before exploring how to extract utility from these phases, it is imperative to address a fundamental constraint of quantum interference: the necessity of uncomputing garbage.

Suppose our quantum circuit $Q_f^\pm$ correctly signs the input state but simultaneously produces residual “garbage” strings $g(x)$ in an array of ancilla qubits. The generated state becomes:

$$|\Psi_{\text{garbage}}\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \otimes |g(x)\rangle$$

In subsequent steps, we will attempt to apply operations that cause the various $|x\rangle$ branches to add or subtract from one another (interference). However, if $g(x) \neq g(y)$, the tensor product components $|x\rangle |g(x)\rangle$ and $|y\rangle |g(y)\rangle$ reside in strictly orthogonal dimensions of the expanded Hilbert space.

The probability amplitudes cannot geometrically overlap, and therefore cannot cancel each other out. The presence of $g(x)$ acts as an implicit, persistent measurement record of the path taken, fundamentally destroying the interference pattern. Thus, the systematic uncomputing of all intermediate garbage (as detailed by Bennett’s trick) is an absolute prerequisite for advanced quantum algorithmic design.

13.5 The Boolean Fourier Transform (Hadamard Transform)

To extract global properties of $f(x)$ from the phase-encoded state $\frac{1}{\sqrt{N}} \sum (-1)^{f(x)} |x\rangle$, we adhere to the overarching paradigm of **Rotate, Compute, Rotate**. The initial rotation created the uniform superposition. The computation signed the amplitudes. The final step mandates a second rotation to induce algorithmic interference. Specifically, we apply a second global layer of Hadamard gates, $H^{\otimes n}$.

To predict the outcome, we must rigorously define the action of $H^{\otimes n}$ on an arbitrary computational basis state $|x\rangle$. Let us consider the effect of $H^{\otimes n}$ on a given target state $|s\rangle$, where $s \in \{0,1\}^n$.

We know the action of a single Hadamard gate:

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ H|1\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \end{aligned}$$

This behavior can be elegantly condensed into a single algebraic rule: $H|x_i\rangle = \frac{1}{\sqrt{2}} \sum_{s_i \in \{0,1\}} (-1)^{x_i s_i} |s_i\rangle$.

Expanding this across the n -fold tensor product:

$$\begin{aligned} H^{\otimes n} |x\rangle &= \left(\frac{1}{\sqrt{2}} \sum_{s_1=0}^1 (-1)^{x_1 s_1} |s_1\rangle \right) \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}} \sum_{s_n=0}^1 (-1)^{x_n s_n} |s_n\rangle \right) \\ &= \frac{1}{\sqrt{2^n}} \sum_{s \in \{0,1\}^n} (-1)^{x_1 s_1 + x_2 s_2 + \cdots + x_n s_n} |s_1 s_2 \dots s_n\rangle \end{aligned}$$

The exponent of the negative sign is exactly the modulo-2 dot product of the bitstrings x and s , effectively evaluating the logical XOR of a designated subset of x 's bits. We formally define this as $s \cdot x$:

$$s \cdot x \equiv \sum_{i=1}^n s_i x_i \pmod{2}$$

Thus, the explicit formula for the n -qubit Boolean Fourier Transform is:

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{N}} \sum_{s \in \{0,1\}^n} (-1)^{s \cdot x} |s\rangle$$

Because the dot product is symmetric ($s \cdot x = x \cdot s$) and the Hadamard transform is its own inverse ($H^{\otimes n} H^{\otimes n} = I$), we can invert the relationship perfectly:

$$|s\rangle = H^{\otimes n} \left(\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} |x\rangle \right) \quad (19)$$

13.6 The Bernstein-Vazirani Algorithm

We now introduce a formal oracle problem proposed by Ethan Bernstein and Umesh Vazirani in 1993, representing one of the earliest rigorous separations between quantum and classical complexity.

The Problem: You are given a black-box circuit (an Oracle) that computes an unknown boolean function $f : \{0,1\}^n \rightarrow \{0,1\}$. You are strictly promised that $f(x)$ is a hidden XOR function, meaning there exists a secret string $s \in \{0,1\}^n$ such that:

$$f(x) = s \cdot x \pmod{2}$$

Your objective is to deduce the secret string s using the minimum possible number of queries to the black box.

Classical Limit: In a classical computing paradigm, evaluating $f(x)$ for a given input x yields a single bit of information regarding the dot product. To deduce the n -bit string s , one must iteratively probe the oracle with canonical basis vectors (e.g., $100\dots 0, 010\dots 0$). This strictly requires exactly n classical queries.

The Quantum Solution: We employ the Rotate-Compute-Rotate protocol.

1. **Initialize:** Prepare the input register in $|0\rangle^{\otimes n}$ and the target ancilla in $|-\rangle$.

2. **Rotate 1:** Apply $H^{\otimes n}$ to the input register to create the uniform superposition: $\frac{1}{\sqrt{N}} \sum_x |x\rangle$.
3. **Compute:** Query the quantum oracle U_f . Utilizing the phase kickback mechanism, the state evolves to:

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle$$

Because we are promised that $f(x) = s \cdot x$, we substitute the definition:

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} |x\rangle$$

4. **Rotate 2:** Apply a final layer of $H^{\otimes n}$. As established rigorously in Equation 19, applying the boolean Fourier transform to this specific phase-encoded state induces massive, deterministic quantum interference. All amplitudes constructively and destructively collapse perfectly into a single basis state:

$$H^{\otimes n} |\Psi\rangle = |s\rangle$$

5. **Measure:** Measure the output register. The readout will display the exact secret string s with 100% probability.

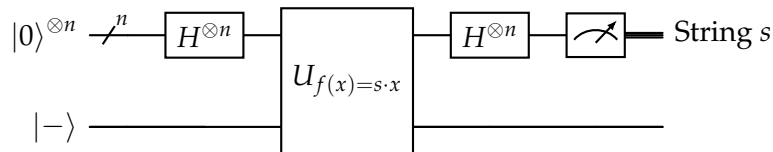


Figure 29: The Bernstein-Vazirani Circuit. By sandwiching the phase-kickback oracle between two layers of Hadamard transforms (Rotate, Compute, Rotate), the global interference pattern deterministically collapses the state onto the secret string s , requiring only a single query.

The Bernstein-Vazirani algorithm fundamentally leverages quantum parallelism and destructive interference to extract a global structural property (n bits of information) in $\mathcal{O}(1)$ queries, vastly outperforming the classical $\mathcal{O}(n)$ bound. This architecture serves as the direct mathematical predecessor to Shor's integer factorization algorithm.

14 Lecture 12: Revealing XOR Patterns and Quantum Fourier Sampling

14.1 Pattern Finding in Implicitly Represented Data

The fundamental power of a quantum computer lies in its unprecedented ability to identify global patterns within implicitly represented datasets. In classical data processing, one routinely searches for periodicities or structural anomalies by applying a Fourier transform to a data vector G of length N . The transform compares the data vector against N distinct pattern vectors, extracting the relative strength (or frequency) of each pattern.

Classically, this operation demands physical computational resources proportional to the size of the dataset. If N is massive—for instance, $N = 2^n$ where n is a parameter like 1000 or 1 000 000—the vector possesses more entries than there are elementary particles in the observable universe. Writing down or processing G explicitly is physically impossible.

A quantum computer, however, can represent a dataset of size $N = 2^n$ using merely n physical qubits. The core advantage of quantum computation is the ability to execute linear transformations—specifically, generalized Fourier transforms—on these exponentially large implicit spaces, extracting the strength of specific patterns in polynomial time $\mathcal{O}(\text{poly}(n))$.

14.2 The Generalized Quantum Fourier Transform

Let $G \in \mathbb{C}^N$ be a normalized data vector. We define a set of N pattern vectors $|\chi_0\rangle, |\chi_1\rangle, \dots, |\chi_{N-1}\rangle$. For this transform to be physically valid in quantum mechanics, these pattern vectors must form an orthonormal basis for $\mathbb{C}^N$.

The transformation from the standard computational basis to this pattern basis is governed by an $N \times N$ unitary matrix U , whose columns are exactly the pattern vectors:

$$U = \begin{pmatrix} | & | & & | \\ |\chi_0\rangle & |\chi_1\rangle & \dots & |\chi_{N-1}\rangle \\ | & | & & | \end{pmatrix}$$

Because the basis vectors are orthonormal, U is strictly unitary ($U^\dagger U = I$). To convert the data vector G into its representation in the pattern basis, we apply the inverse transformation, $U^\dagger$.

When G passes through this linear transformation, the resulting state yields the Fourier coefficients (the “strengths”) of each pattern. We denote the s -th Fourier coefficient as $\hat{G}_s$:

$$\hat{G}_s = \langle \chi_s | G \rangle$$

By the definition of the inner product, this scalar evaluates how strongly the pattern $|\chi_s\rangle$ is correlated with the data vector G . If we measure the quantum state after applying $U^\dagger$, the probability of observing the classical index s corresponding to a specific pattern is precisely given by the Born rule:

$$P(s) = |\hat{G}_s|^2 = |\langle \chi_s | G \rangle|^2$$

14.3 Functions as Quantum States and Normalization

To interface classical computer science with quantum mechanics, we must map boolean functions to quantum state vectors. Consider a boolean function $F : \{0, 1\}^n \rightarrow \{0, 1\}$. Evaluating its truth table generates a sequence of $N = 2^n$ bits.

To encode this table as a valid quantum state, we first map the classical binary outputs $\{0, 1\}$ to the phase parities $\{+1, -1\}$ using the transformation $f(x) = (-1)^{F(x)}$. We treat this sequence of N signs as the amplitudes of a complex vector. However, to satisfy the axiomatic requirement of quantum mechanics that the sum of squared magnitudes equals 1, we must introduce a global normalization constant of $\frac{1}{\sqrt{N}}$.

We introduce the formal notation $|f\rangle$ to denote the quantum state explicitly encoding the truth table of the function:

$$|f\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} f(x) |x\rangle$$

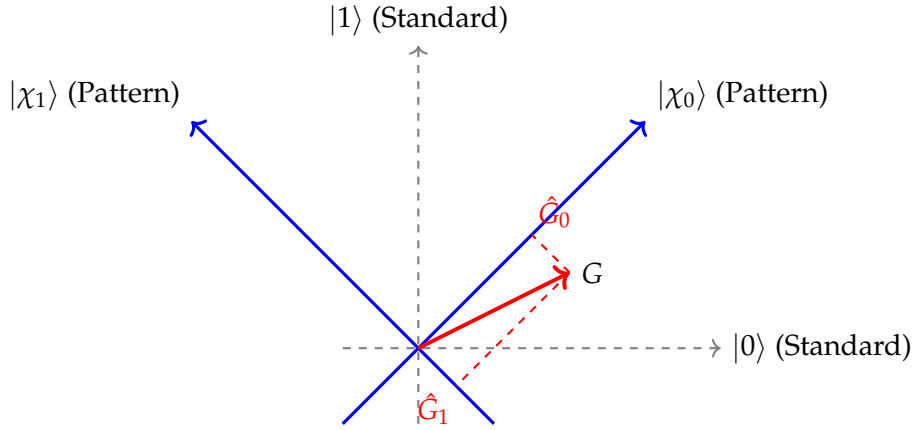


Figure 30: Geometric interpretation of the quantum Fourier transform. The initial data vector G is projected onto a new orthonormal basis formed by the pattern vectors $|\chi_s\rangle$. The squared magnitude of the resulting coordinates ($\hat{G}_s$) dictates the measurement probabilities.

More generally, for any complex-valued function $G(x)$, it constitutes a valid quantum state if its expected squared magnitude over the boolean hypercube is identically 1:

$$\frac{1}{N} \sum_{x \in \{0,1\}^n} |G(x)|^2 = \mathbb{E}_x [|G(x)|^2] = 1$$

14.4 Fourier Coefficients as Statistical Correlation

Using this function-state isomorphism, we can reformulate the Fourier coefficient $\hat{G}_s$ as a statistical correlation. Expanding the inner product $\langle \chi_s | G \rangle$:

$$\begin{aligned} \hat{G}_s &= \left(\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} \chi_s^*(x) \langle x | \right) \left(\frac{1}{\sqrt{N}} \sum_{y \in \{0,1\}^n} G(y) |y\rangle \right) \\ &= \frac{1}{N} \sum_{x \in \{0,1\}^n} \chi_s^*(x) G(x) \\ &= \mathbb{E}_x [\chi_s^*(x) G(x)] \end{aligned}$$

When both the pattern function $\chi_s(x)$ and the data function $G(x)$ strictly take boolean parity values in $\{+1, -1\}$, their product $\chi_s(x)G(x)$ is $+1$ where the functions agree, and -1 where they disagree. Thus, the Fourier coefficient reduces to a straightforward statistical discrepancy:

$$\hat{G}_s = (\text{Fraction where } G(x) = \chi_s(x)) - (\text{Fraction where } G(x) \neq \chi_s(x))$$

The coefficient $\hat{G}_s \in [-1, 1]$ directly measures the geometric alignment or classical correlation between the implicit dataset and the target pattern.

14.5 The Boolean Fourier Transform and XOR Functions

While any orthonormal basis yields a mathematically valid transform, quantum algorithmic design focuses on bases that are physically efficiently implementable. The most foundational of these is the **Boolean Fourier Transform**, which utilizes parity (XOR) functions as its pattern vectors.

For every boolean string $s \in \{0,1\}^n$, we define the corresponding XOR function pattern $\chi_s(x)$ as:

$$\chi_s(x) = (-1)^{s \cdot x}$$

where $s \cdot x$ is the bitwise dot product modulo 2:

$$s \cdot x = \sum_{i=1}^n s_i x_i \pmod{2}$$

These pattern vectors exhibit an extraordinarily rare and mathematically beautiful property: every single entry in the vector (ignoring the global $\frac{1}{\sqrt{N}}$ normalization) is exactly $+1$ or -1 . Generic orthonormal bases in high-dimensional spaces generally comprise complex irrational numbers; the fact that the XOR basis is purely boolean allows for perfect alignment with classical digital functions.

14.6 The Hadamard Transform Matrix

The transformation matrix U whose columns correspond to these XOR pattern vectors is exactly the n -qubit Hadamard transformation, $H^{\otimes n}$.

Let us rigorously analyze the structure for $n = 1$ ($N = 2$) and $n = 2$ ($N = 4$):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H^{\otimes 2} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Notice that the rows and columns are indexed by the boolean strings x and s . The generic matrix entry at row x and column s is perfectly described by our XOR formula:

$$(H^{\otimes n})_{x,s} = \frac{1}{\sqrt{2^n}} (-1)^{x \cdot s}$$

Because the dot product modulo 2 is symmetric ($x \cdot s = s \cdot x$), the matrix $H^{\otimes n}$ is perfectly symmetric. As the Hadamard gates are unitary and Hermitian, the transform is its own inverse: $H^{\otimes n} H^{\otimes n} = I$.

14.7 The Quantum Fourier Sampling Paradigm

We now possess all mathematical components to construct the canonical quantum algorithmic template: **Fourier Sampling**. This paradigm encapsulates the execution of algorithms ranging from Bernstein-Vazirani to Simon's and Shor's algorithm.



Figure 31: The Quantum Fourier Sampling Paradigm. The sequence rigorously follows the *Rotate* ($H^{\otimes n}$), *Compute* ($U_F^{\pm}$), *Rotate* ($H^{\otimes n}$) structure. The intermediate oracle performs a phase-kickback sign implementation, mapping the amplitudes to the functional outputs.

The algebraic trace of the algorithm proceeds as follows:

1. **Initialization:** Prepare the n -qubit input register in the ground state $|0\rangle^{\otimes n}$.
2. **First Rotation:** Apply $H^{\otimes n}$ to generate the uniform superposition over all boolean strings:

$$|\Psi_1\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle$$

3. **Computation (Phase Kickback):** Query the sign-implementing oracle $U_F^\pm$. This dynamically loads the implicit dataset into the probability amplitudes:

$$|\Psi_2\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle \equiv |f\rangle$$

4. **Second Rotation:** Apply the Boolean Fourier Transform ($H^{\otimes n}$) to project the functional state into the XOR pattern basis.

$$|\Psi_3\rangle = \sum_{s \in \{0,1\}^n} \hat{f}_s |s\rangle$$

5. **Measurement:** Measure the register. The classical string s is observed with probability precisely $P(s) = |\hat{f}_s|^2$.

14.8 Oracle Complexities and Quantum Separations

This paradigm allows us to formalize the exponential separation in computational power between classical models (Deterministic P, Randomized BPP) and the quantum model (BQP) through a series of oracle problems.

1. The Bernstein-Vazirani Algorithm (1992): You are provided a black-box circuit computing $F(x)$ with the strict structural promise that $F(x) = s \cdot x \pmod{2}$ for some hidden secret string s . In our Fourier language, the function vector $|f\rangle$ is identically equal to the specific pattern vector $|\chi_s\rangle$. When passed through the Fourier Sampling circuit, the final state evaluates to:

$$\hat{f}_{s'} = \langle \chi_{s'} | f \rangle = \langle \chi_{s'} | \chi_s \rangle = \delta_{s,s'}$$

Due to orthonormality, the amplitude is strictly 1 for the target string s , and 0 everywhere else. A single measurement yields the secret string s with 100% determinism.

- Classical Deterministic Queries: n
- Quantum Queries: 1

2. The Deutsch-Jozsa Algorithm (1992): You are given a function $F(x)$ promised to be either *Constant* (outputs exactly the same bit for all x) or *Balanced* (outputs 0 for half the inputs, and 1 for the other half). You must determine which. Passing this through the Fourier Sampling paradigm and observing strictly the $|0\rangle^{\otimes n}$ output (the $s = 0 \dots 0$ pattern): If the function is constant, the strength of the empty pattern χ_0 is absolute (100% amplitude). The algorithm outputs $0 \dots 0$ deterministically. If the function is balanced, its correlation with the constant vector is exactly 0, guaranteeing the readout will *never* be $0 \dots 0$.

- Classical Deterministic Queries: $2^{n-1} + 1$ (Exponential worst-case)
- Classical Randomized (BPP) Queries: $\mathcal{O}(1)$
- Quantum Queries: 1

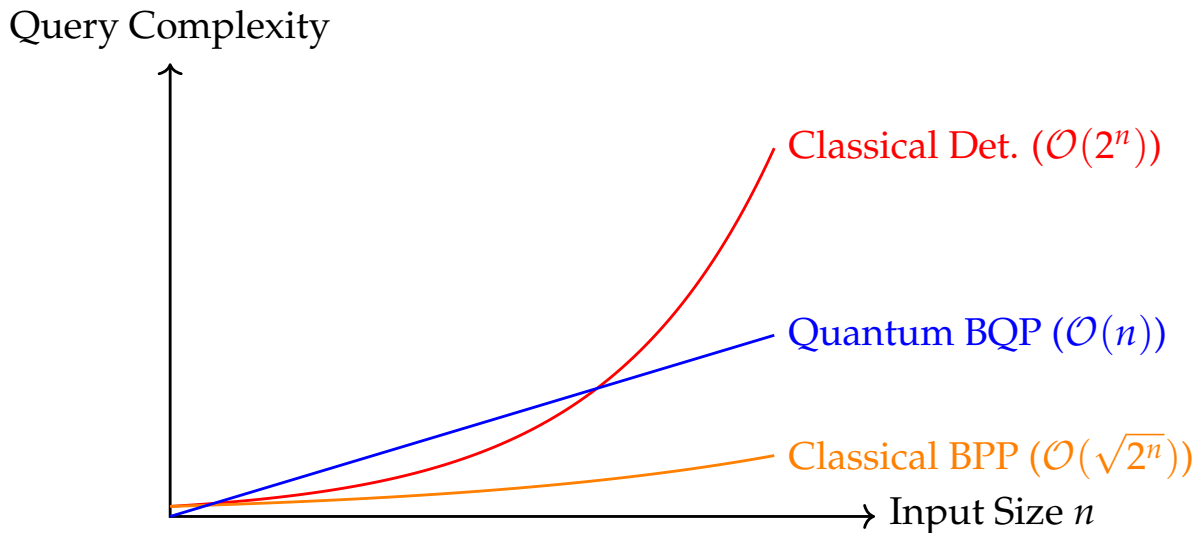


Figure 32: Asymptotic query complexity for Simon's Problem. This algorithm provides the first rigorous proof of an exponential separation between the probabilistic classical complexity class (BPP) and the bounded-error quantum complexity class (BQP).

3. Simon's Algorithm (1994): While Deutsch-Jozsa offers an exponential separation against deterministic algorithms, classical computers seamlessly solve it using randomness. Simon's algorithm presents a function $F : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with a hidden periodic XOR structure (a secret string s such that $F(x) = F(x \oplus s)$). Finding s is the first problem to definitively conquer both deterministic and probabilistic classical bounds.

The exponential speedup demonstrated by Simon's implementation of the Fourier Sampling Paradigm directly inspired Peter Shor to generalize the method from the Boolean field to integer arithmetic, leading to the efficient factorization of prime products and the subsequent vulnerability of RSA cryptography.

15 Lecture 13: Simon's Algorithm and Exponential Quantum Speedups

15.1 The Fourier Sampling Paradigm and Oracle Models

Simon's Algorithm, developed by Dan Simon, represents a watershed moment in quantum computer science. It was the first rigorous demonstration of an exponential speedup by a quantum computer over classical randomized algorithms (the BPP complexity class). Furthermore, it directly established the algorithmic architecture—the "Rotate, Compute, Rotate" Fourier sampling paradigm—that Peter Shor would later adapt to efficiently factor integers.

Unlike standard computational problems, Simon's problem is formulated within the *black-box query model* (or oracle model). We are provided access to a quantum circuit Q_F that implements an unknown function $F : \{0, 1\}^n \rightarrow \{0, 1\}^m$. We treat this circuit as an opaque hardware module; we cannot reverse-engineer its internal gates. Our objective is to determine a hidden global property of F using the minimum possible number of queries to Q_F .

Because F maps n bits to m bits (where $m \geq n - 1$), it is pedagogically useful to interpret the m -bit outputs not as raw integers, but as an exponentially large palette of *colors*. Thus, the function F essentially assigns a specific color to each vertex of the n -dimensional Boolean hypercube $\mathbb{F}_2^n$.

15.2 Simon's Promise: Periodicity on the Boolean Cube

The black-box function F is not completely arbitrary; we are given a strict mathematical promise that F is **periodic** with a secret period $L \in \{0, 1\}^n$, where $L \neq 0^n$.

In the vector space $\mathbb{F}_2^n$ (where addition is bitwise XOR, denoted by $\oplus$), the formal definition of L -periodicity requires that for all $x \in \{0, 1\}^n$:

$$F(x \oplus L) = F(x)$$

Furthermore, we append a strict uniqueness condition to this definition: F must map elements to the same color *if and only if* they are separated by the vector L . Algebraically, this means:

$$F(x) = F(y) \iff x \oplus y \in \{0^n, L\}$$

Because every input x is paired with exactly one distinct partner $x \oplus L$ that shares the same color, the function divides the 2^n possible inputs into exactly 2^{n-1} disjoint pairs. Consequently, the range of the function contains exactly 2^{n-1} distinct colors, which necessitates that the output dimension m must be at least $n - 1$.

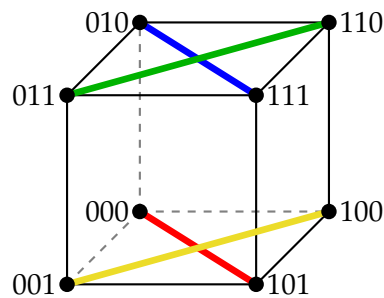


Figure 33: Visual representation of an L -periodic function on the Boolean hypercube for $n = 3$, with the secret period $L = 101$. The vertices are paired across the diagonals of the respective faces, dictating that $F(000) = F(101) = \text{Red}$, $F(001) = F(100) = \text{Yellow}$, etc.

15.3 Classical Complexity Limits

Before deploying quantum mechanics, we must strictly bound the classical difficulty of finding L . Since the oracle is a black box, a classical algorithm (deterministic or randomized) must submit classical query strings x_i and observe the output colors C_i .

If the algorithm finds a collision—two distinct inputs $x_i \neq x_j$ such that $F(x_i) = F(x_j)$ —it succeeds immediately, as the periodicity promise dictates that $x_i \oplus x_j = L$. However, if it observes k distinct colors, it has merely ruled out $\binom{k}{2}$ possible values for L . Because the space of possible periods is $2^n - 1$, the Birthday Paradox governs the expected number of queries to find a collision among uniformly drawn pairs.

To achieve a high probability of finding a collision, the number of evaluated pairs $\mathcal{O}(T^2)$ must scale with the size of the target space $\mathcal{O}(2^n)$. Therefore, any classical algorithm requires an exponential number of queries:

$$T_{\text{classical}} = \Omega\left(\sqrt{2^n}\right) = \Omega\left(2^{n/2}\right)$$

For physically macroscopic instances (e.g., $n = 1000$), 2^{500} queries exceed the computational capacity of the universe, rendering the problem classically intractable.

15.4 The Quantum Algorithm: Data Loading and Partial Measurement

The quantum solution requires an input register of n qubits and an output register of m qubits. We initialize the system in the ground state:

$$|\Psi_0\rangle = |0\rangle^{\otimes n} \otimes |0\rangle^{\otimes m}$$

Applying the first rotation (the Boolean Fourier Transform) via a layer of Hadamard gates on the input register yields the uniform superposition over all possible classical inputs:

$$|\Psi_1\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \right) \otimes |0\rangle^{\otimes m}$$

We now compute the function by passing the state through the Oracle Q_F . Unlike the phase-kickback trick used in Bernstein-Vazirani, F is a multi-bit function, so we entangle the input directly with the target register:

$$|\Psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |F(x)\rangle$$

This single operation essentially evaluates all 2^n inputs simultaneously, loading the entire implicit dataset into the massive state vector.

To isolate the periodicity, we conceptually perform a partial measurement exclusively on the m -qubit output register. The measurement apparatus will read out a specific classical color $C^* \in \{0,1\}^m$. By the Born rule, the probability of obtaining any specific color C^* is the sum of the squared amplitudes of the corresponding basis states. Since C^* is generated by exactly two inputs (x^* and $x^* \oplus L$), the probability is:

$$P(C^*) = \left| \frac{1}{\sqrt{2^n}} \right|^2 + \left| \frac{1}{\sqrt{2^n}} \right|^2 = \frac{2}{2^n} = \frac{1}{2^{n-1}}$$

This uniform probability reflects that we have no control over which color we observe. However, upon observing C^* , the global wave-function irreversibly collapses. The n -qubit input register is stripped of all terms inconsistent with the observation and renormalized, leaving an equal superposition of exactly two states:

$$|\Psi_3\rangle = \frac{1}{\sqrt{2}} |x^*\rangle + \frac{1}{\sqrt{2}} |x^* \oplus L\rangle$$

While this state encapsulates the secret L , measuring it now is useless. A measurement would simply yield either x^* or $x^* \oplus L$, independently uniformly distributed. If we rerun the algorithm, we will observe a *different* random color, collapsing to a completely different disjoint pair, generating zero mutual information about L .

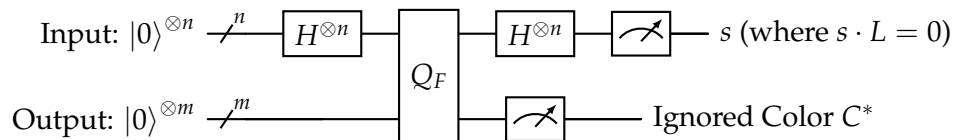


Figure 34: The Circuit for Simon's Algorithm. Following the Principle of Deferred Measurement, the partial measurement of the output register can technically be omitted without altering the statistical distribution of the input register's final measurement. The core logic relies strictly on the terminal $H^{\otimes n}$ rotation inducing selective interference.

15.5 The Final Rotation: Inducing Interference

To extract the invariant global property L independent of the random local shift x^* , we execute the final phase of the Fourier sampling paradigm. We apply the n -qubit Boolean Fourier Transform ($H^{\otimes n}$) to our collapsed two-state superposition $|\Psi_3\rangle$.

Recall the algebraic action of the Hadamard transform on a basis state $|x\rangle$:

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{s \in \{0,1\}^n} (-1)^{x \cdot s} |s\rangle$$

Applying this linearly to $|\Psi_3\rangle$:

$$\begin{aligned} |\Psi_4\rangle &= H^{\otimes n} \left(\frac{1}{\sqrt{2}} |x^*\rangle + \frac{1}{\sqrt{2}} |x^* \oplus L\rangle \right) \\ &= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2^n}} \sum_{s \in \{0,1\}^n} (-1)^{x^* \cdot s} |s\rangle + \frac{1}{\sqrt{2^n}} \sum_{s \in \{0,1\}^n} (-1)^{(x^* \oplus L) \cdot s} |s\rangle \right) \end{aligned}$$

By the distributive property of the modulo-2 dot product, $(x^* \oplus L) \cdot s = (x^* \cdot s) \oplus (L \cdot s)$. This allows us to factor out the common phase:

$$\begin{aligned} |\Psi_4\rangle &= \frac{1}{\sqrt{2^{n+1}}} \sum_{s \in \{0,1\}^n} \left[(-1)^{x^* \cdot s} + (-1)^{(x^* \cdot s) \oplus (L \cdot s)} \right] |s\rangle \\ &= \frac{1}{\sqrt{2^{n+1}}} \sum_{s \in \{0,1\}^n} (-1)^{x^* \cdot s} \left(1 + (-1)^{L \cdot s} \right) |s\rangle \end{aligned}$$

This equation governs the destructive and constructive interference patterns of the algorithm. We evaluate the parenthetical term based on the boolean value of $L \cdot s$:

- **Case 1** ($L \cdot s = 1$): The term evaluates to $1 + (-1)^1 = 0$. The probability amplitude strictly annihilates due to perfect destructive interference.
- **Case 2** ($L \cdot s = 0$): The term evaluates to $1 + (-1)^0 = 2$. The probability amplitude constructively reinforces.

Substituting these conditions, the un-annihilated states remain:

$$|\Psi_4\rangle = \frac{2}{\sqrt{2^{n+1}}} \sum_{\substack{s \in \{0,1\}^n \\ L \cdot s = 0}} (-1)^{x^* \cdot s} |s\rangle = \sqrt{\frac{2}{2^n}} \sum_{\substack{s \in \{0,1\}^n \\ L \cdot s = 0}} (-1)^{x^* \cdot s} |s\rangle$$

Because $L \neq 0^n$, the orthogonal complement space $L^\perp = \{s \mid L \cdot s = 0 \pmod{2}\}$ is an $(n-1)$ -dimensional subspace containing exactly 2^{n-1} vectors. Notice that the measurement probability for any valid s is $|\sqrt{2/2^n}|^2 = 2/2^n = 1/2^{n-1}$, confirming that the state is properly normalized.

When we measure this final state, the problematic random shift x^* is absorbed strictly into the unobservable complex phase $(-1)^{x^* \cdot s}$. The measurement outcome is a uniformly random vector s sampled strictly from the orthogonal complement of L . We have effectively obtained a guaranteed, true linear equation concerning the secret bits of L :

$$s \cdot L \equiv s_1 L_1 \oplus s_2 L_2 \oplus \cdots \oplus s_n L_n = 0 \pmod{2}$$

15.6 Classical Post-Processing and Total Complexity

A single execution of Simon's algorithm yields one linear equation. Because L is an n -bit vector, uniquely resolving it strictly requires $n - 1$ linearly independent equations (since $L = 0^n$ is already ruled out by the promise). We therefore repeat the quantum subroutine $\mathcal{O}(n)$ times, obtaining a system of equations:

$$\begin{pmatrix} s_1^{(1)} & s_2^{(1)} & \dots & s_n^{(1)} \\ s_1^{(2)} & s_2^{(2)} & \dots & s_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ s_1^{(k)} & s_2^{(k)} & \dots & s_n^{(k)} \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \\ \vdots \\ L_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \pmod{2}$$

We must verify the probability that drawing $n - 1$ vectors from $L^\perp$ yields a linearly independent set. The first vector $s^{(1)}$ is linearly independent as long as it is not the zero vector, occurring with probability $1 - \frac{1}{2^{n-1}}$. The second vector is independent if it does not fall into the 1-dimensional subspace spanned by the first, occurring with probability $1 - \frac{2}{2^{n-1}}$. Extending this logic, the probability of obtaining a full rank set of $n - 1$ equations in $n - 1$ draws is:

$$P(\text{Full Rank}) = \prod_{i=0}^{n-2} \left(1 - \frac{2^i}{2^{n-1}}\right)$$

As $n \rightarrow \infty$, this infinite product rigorously evaluates to approximately 0.288. By bounding the lower terms, we analytically establish:

$$P(\text{Full Rank}) \geq \frac{1}{4}$$

Because the success probability of obtaining the independent system is bounded firmly away from zero ($\geq 25\%$), we simply execute the batch of queries, perform classical Gaussian elimination in $\mathcal{O}(n^3)$ time, and check the validity of L . If the rank is deficient, we rerun the procedure. The expected number of required quantum oracle queries scales linearly:

$$\mathbb{E}[T_{\text{quantum}}] \leq 4n = \mathcal{O}(n)$$

Simon's algorithm provides an incontrovertible mathematical proof that the bounded-error quantum polynomial complexity class (BQP) strictly contains computational capabilities entirely impossible for classical bounded-error probabilistic algorithms (BPP).

16 Lecture 14: The Fourier Transform over $\mathbb{Z}_N$

16.1 Transitioning from Boolean Cubes to Modular Arithmetic

In our previous analyses, we thoroughly explored the Boolean Fourier Transform (the Hadamard transform) and its application in Simon's algorithm. That framework operated over the vector space $\mathbb{F}_2^n$, where the foundational operation was bitwise addition modulo 2 (the logical XOR operation). Simon's algorithm demonstrated how a quantum computer could identify a hidden period L in a function $f(x) = f(x \oplus L)$ using exponentially fewer queries than any classical probabilistic algorithm ($\mathcal{O}(n)$ vs. $\mathcal{O}(2^{n/2})$).

However, many of the most profound problems in computer science and number theory—most notably integer factorization and the breaking of RSA cryptography—do not rely on bitwise XOR. Instead, they rely on standard modular arithmetic. Shor's algorithm, for

instance, requires identifying the period L of numerical exponential functions, such as $F(x) = a^x \pmod{M}$, where the periodicity obeys standard integer addition:

$$F(x) = F(x + L) = F(x + 2L) = \dots \pmod{M}$$

To identify global periodic structures over integer addition, we must upgrade our quantum tools from the Boolean Fourier Transform to the discrete Fourier transform over the cyclic group $\mathbb{Z}_N$ (the integers modulo N). Throughout this chapter, N will denote an exponentially large dimension, typically $N = 2^n$, where n represents the physical number of qubits in our register (e.g., $n = 1000$ implies $N \approx 10^{300}$).

16.2 Functions as Quantum States and Normalization

We study functions mapping the integers modulo N to the complex plane: $G : \mathbb{Z}_N \rightarrow \mathbb{C}$. In quantum computation, we natively represent such a discrete function as a quantum state vector of length N , whose amplitudes correspond to the function's truth table or data sequence.

Using Dirac notation, the function G is embedded into a state vector $|G\rangle$ by taking a coherent superposition over all N basis states, indexed by $x \in \{0, 1, \dots, N-1\}$:

$$|G\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} G(x) |x\rangle$$

For $|G\rangle$ to be a physically valid quantum state, the sum of its squared amplitude magnitudes must rigorously equal 1. This enforces the normalization constraint:

$$\frac{1}{N} \sum_{x=0}^{N-1} |G(x)|^2 = \mathbb{E}_x [|G(x)|^2] = 1$$

16.3 Characters of $\mathbb{Z}_N$: The Roots of Unity

The discrete Fourier transform fundamentally decomposes an arbitrary function G into a linear combination of orthonormal “pattern functions”. In algebraic group theory, these pattern functions are known as *characters*.

For the Boolean case, the characters were the parity functions $\chi_s(x) = (-1)^{s \cdot x}$, which satisfied the structural homomorphism:

$$\chi_s(x \oplus y) = \chi_s(x) \chi_s(y) \tag{20}$$

We require a new set of character functions $\chi_s(x)$ for $\mathbb{Z}_N$ that satisfy the additive equivalent of Equation 20:

$$\chi_s(x + y) = \chi_s(x) \chi_s(y)$$

This homomorphic property forces the answer upon us. Setting $y = 0$ immediately reveals that $\chi_s(0) = 1$. Furthermore, sequentially expanding $\chi_s(N \cdot x)$ under this rule yields:

$$\chi_s(x + x + \dots + x) = (\chi_s(x))^N$$

Since we are operating modulo N , $N \cdot x \equiv 0 \pmod{N}$. Therefore:

$$\chi_s(0) = (\chi_s(x))^N \implies (\chi_s(x))^N = 1$$

This rigorously proves that the values of the character functions must be drawn strictly from the N -th roots of unity in the complex plane.

We define the primitive N -th root of unity, ω_N , as:

$$\omega_N = e^{2\pi i/N} = \cos\left(\frac{2\pi}{N}\right) + i \sin\left(\frac{2\pi}{N}\right)$$

For each integer $s \in \{0, \dots, N-1\}$, the corresponding pattern function $\chi_s(x)$ is defined as:

$$\chi_s(x) = \omega_N^{sx}$$

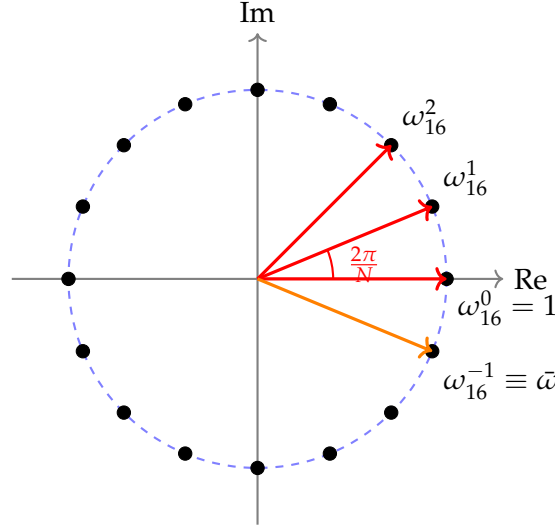


Figure 35: The N -th roots of unity in the complex plane for $N = 16$. The primitive root ω_{16} generates the cyclic group. The discrete Fourier transform maps standard basis vectors into superpositions constructed from these complex phases.

Note that if we examine the real part of these characters as x varies, $\text{Re}(\chi_s(x)) = \cos(2\pi sx/N)$, they trace out discrete cosine waves of increasing frequency, providing a direct mathematical analog to classical continuous harmonic analysis.

16.4 Orthonormality of the Character Basis

To utilize these pattern functions as a basis for quantum transformations, we must prove they are strictly orthonormal. The inner product between two arbitrary character states $|\chi_t\rangle$ and $|\chi_s\rangle$ is:

$$\langle \chi_t | \chi_s \rangle = \frac{1}{N} \sum_{x=0}^{N-1} \chi_t(x)^* \chi_s(x)$$

Using the property of complex conjugation for roots of unity (ω^{-tx}), we combine the terms:

$$\langle \chi_t | \chi_s \rangle = \frac{1}{N} \sum_{x=0}^{N-1} \omega_N^{-tx} \omega_N^{sx} = \frac{1}{N} \sum_{x=0}^{N-1} \omega_N^{(s-t)x}$$

Case 1 ($s = t$): The exponent is 0, making every term exactly $\omega_N^0 = 1$. The sum evaluates to $\frac{1}{N}(N) = 1$. The vectors are properly normalized unit states.

Case 2 ($s \neq t$): Let $r = (s - t) \not\equiv 0 \pmod{N}$. The summation becomes a geometric series:

$$\sum_{x=0}^{N-1} (\omega_N^r)^x = \frac{1 - (\omega_N^r)^N}{1 - \omega_N^r}$$

Because $(\omega_N^r)^N = 1^r = 1$, the numerator evaluates strictly to 0. Thus, $\langle \chi_t | \chi_s \rangle = 0$, guaranteeing that all N character vectors are mutually orthogonal.

16.5 The Discrete Fourier Transform Unitary Matrix

Because the vectors $|\chi_s\rangle$ form an orthonormal basis, stacking them as columns generates an $N \times N$ unitary matrix. The transformation mapping the standard computational basis to this character basis (and vice versa) defines the Quantum Fourier Transform (QFT).

The matrix elements of the direct Quantum Fourier Transform are conventionally defined as:

$$(U_{\text{QFT}})_{s,x} = \frac{1}{\sqrt{N}} \omega_N^{sx}$$

Its inverse (the inverse QFT, which is mathematically identical to its conjugate transpose $U_{\text{QFT}}^\dagger$) has matrix elements:

$$(U_{\text{QFT}}^\dagger)_{s,x} = \frac{1}{\sqrt{N}} \omega_N^{-sx}$$

Applying $U_{\text{QFT}}^\dagger$ to an initial data vector $|G\rangle$ projects it into the Fourier domain, yielding the spectral coefficients $\hat{G}_s$. The paramount property of this transformation is that if the sequence $G(x)$ exhibits a periodic spike structure with period L , its Fourier transform $\hat{G}_s$ will exhibit highly constructive, observable interference spikes strictly at multiples of N/L .

16.6 Circuit Construction: The Exponential Speedup

In classical computing, multiplying an arbitrary vector by an $N \times N$ matrix requires $\mathcal{O}(N^2)$ operations. The celebrated Fast Fourier Transform (FFT) algorithm optimizes this to $\mathcal{O}(N \log N)$. However, when $N = 2^n$ (e.g., $N = 2^{1000}$), even the FFT is fundamentally blocked by the boundaries of physical reality.

Remarkably, whenever a $2^n \times 2^n$ unitary matrix possesses a purely algebraic closed-form definition—like ω_N^{sx} —it frequently permits exponential compilation down to a sequence of basic quantum gates. We will demonstrate that a quantum computer can execute the QFT using only $\mathcal{O}(n^2)$ one- and two-qubit gates, achieving an exponential speedup.

For explicit derivation, let $n = 4$ qubits ($N = 16$). We apply the Inverse QFT to a computational basis state $|x\rangle$. To avoid cluttered notation, we define $\bar{\omega} \equiv \omega_{16}^{-1}$. The target state is:

$$U_{\text{IQFT}} |x\rangle = \frac{1}{\sqrt{16}} \sum_{s=0}^{15} \bar{\omega}^{sx} |s\rangle$$

We write the summation index s in its binary representation $s = 8s_3 + 4s_2 + 2s_1 + s_0$, where $s_i \in \{0, 1\}$. This allows us to strictly factor the massive summation into an unentangled tensor product of four independent qubits:

$$U_{\text{IQFT}} |x\rangle = \frac{1}{4} \left(\sum_{s_3=0}^1 \bar{\omega}^{8s_3x} |s_3\rangle \right) \otimes \left(\sum_{s_2=0}^1 \bar{\omega}^{4s_2x} |s_2\rangle \right) \otimes \left(\sum_{s_1=0}^1 \bar{\omega}^{2s_1x} |s_1\rangle \right) \otimes \left(\sum_{s_0=0}^1 \bar{\omega}^{s_0x} |s_0\rangle \right)$$

Evaluating the summations over $\{0, 1\}$, we observe the explicit structure of the unentangled output register:

$$|\Psi_{\text{out}}\rangle = \frac{1}{4} \underbrace{(|0\rangle + \bar{\omega}^{8x} |1\rangle)}_{\text{Qubit 3}} \otimes \underbrace{(|0\rangle + \bar{\omega}^{4x} |1\rangle)}_{\text{Qubit 2}} \otimes \underbrace{(|0\rangle + \bar{\omega}^{2x} |1\rangle)}_{\text{Qubit 1}} \otimes \underbrace{(|0\rangle + \bar{\omega}^x |1\rangle)}_{\text{Qubit 0}} \quad (21)$$

We must construct a local circuit to generate these highly specific phase factors. We substitute the binary representation of the input $x = 8x_3 + 4x_2 + 2x_1 + x_0$:

- **Qubit 3:** The phase is $\bar{\omega}^{8x} = \bar{\omega}^{64x_3+32x_2+16x_1+8x_0}$. Because $\bar{\omega}^{16} = 1$, all higher multiples vanish. The phase reduces identically to $\bar{\omega}^{8x_0}$. Since $\bar{\omega}^8 = -1$, this is exactly $(-1)^{x_0}$. The target state is $|0\rangle + (-1)^{x_0}|1\rangle$. This is cleanly achieved by applying a single Hadamard gate to the input bit x_0 .
- **Qubit 2:** The phase is $\bar{\omega}^{4x} = \bar{\omega}^{32x_3+16x_2+8x_1+4x_0} = \bar{\omega}^{8x_1}\bar{\omega}^{4x_0} = (-1)^{x_1}\bar{\omega}^{4x_0}$. We achieve this by applying a Hadamard gate to x_1 (generating the -1 parity), followed by a controlled phase rotation $\bar{\omega}^4$ governed exclusively by x_0 .
- **Qubit 1:** The phase $\bar{\omega}^{2x}$ depends on x_2 (via H), x_1 (via controlled- $\bar{\omega}^4$), and x_0 (via controlled- $\bar{\omega}^2$).
- **Qubit 0:** The phase $\bar{\omega}^x$ depends on all four input bits.

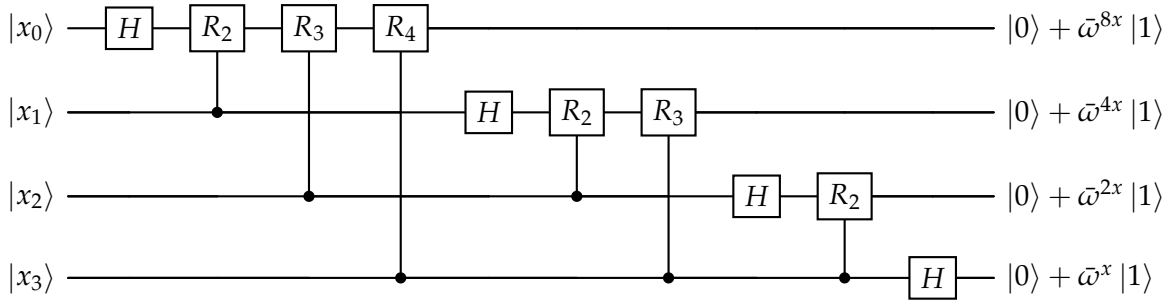


Figure 36: The cascaded quantum circuit for the 4-qubit Inverse QFT. The operator R_k denotes a controlled phase rotation by an angle $2\pi/2^k$. The output qubits strictly emerge in reverse bit-significance order.

The structure clearly generalizes to n qubits. The first wire demands 1 gate; the second requires 2; the n -th requires n . The total gate complexity is mathematically exactly the arithmetic progression:

$$\text{Total Gates} = \sum_{k=1}^n k = \frac{n(n+1)}{2} = \mathcal{O}(n^2)$$

16.7 Coppersmith's Approximate QFT and Physical Reality

A critical hardware objection arises at extreme computational scales. If $n = 1000$, the phase gate required between the first and last wire is a controlled rotation of $\bar{\omega}^1 = e^{-2\pi i/2^{1000}}$. This angle is physically infinitesimal. It corresponds to adjusting the thickness of a quartz phase-shifter by $\sim 10^{-300}$ meters, or pulsing a laser for 10^{-300} seconds—thresholds utterly obscured by fundamental Planck limits and quantum noise.

Don Coppersmith resolved this barrier by rigorously defining the **Approximate Quantum Fourier Transform**. He observed that dropping all controlled phase gates where the rotational angle θ falls below a specified macroscopic threshold ϵ introduces only bounded operator error.

By strategically truncating interactions that span beyond $\mathcal{O}(\log(n/\epsilon))$ adjacent wires, the algorithm computes a unitary operator that is bounded ϵ -close to the ideal theoretical QFT matrix in the operator norm. Consequently, the success probability of downstream applications—such as Shor's algorithm—is perturbed by at most $\mathcal{O}(\epsilon)$.

This brilliant truncation reduces the operational circuit complexity from strictly $\mathcal{O}(n^2)$ down to a near-linear $\mathcal{O}(n \log(n/\epsilon))$, validating that exponential Fourier processing over $\mathbb{Z}_N$ is not merely a theoretical curiosity, but a physically plausible architecture.